

Chapter 27 Check

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27 Supersymmetric Gauge Theories

The successful theories of strong, weak, and electromagnetic interactions described in the first two volumes are all gauge theories. In order to see how simple supersymmetry may make contact with reality, we must therefore consider how to construct actions that satisfy both supersymmetry and gauge invariance [1].

27.1 Gauge-Invariant Actions for Chiral Superfields

Consider a set of Abelian or non-Abelian gauge transformations that leave the supersymmetry generator Q invariant. (For simple supersymmetry there is just one Majorana spinor supersymmetry generator, which can only furnish a trivial representation of any semi-simple gauge group.) Each component field in a supermultiplet must transform in the same way under such gauge transformations. In particular, for a left-chiral superfield we have

$$\begin{aligned}
\phi_n(x) &\longrightarrow \sum_m \left[\exp \left(i \sum_A t_A \Lambda^A(x) \right) \right]_{nm} \phi_m(x), \\
\psi_{nL}(x) &\longrightarrow \sum_m \left[\exp \left(i \sum_A t_A \Lambda^A(x) \right) \right]_{nm} \psi_{mL}(x), \\
\mathcal{F}_n(x) &\longrightarrow \sum_m \left[\exp \left(i \sum_A t_A \Lambda^A(x) \right) \right]_{nm} \mathcal{F}_m(x),
\end{aligned} \tag{27.1.1}$$

where t_A are Hermitian matrices representing the generators of the gauge algebra, and $\Lambda^A(x)$ are real functions of x^μ that parameterize a finite gauge transformation. (We are using the same notation for gauge transformations as in Section 15.1, except that in order to

avoid confusion with Dirac indices we label the gauge generators and gauge transformation parameters with the letters A, B , etc. instead of α, β , etc.)

The left-chiral superfield (??) involves derivatives of some of the component fields, so its transformation is more complicated than Eq. (27.1.1). However, Eq. (??) shows that the superfield does not involve derivatives if expressed in terms of θ_L and the variable x_+ defined by Eq. (??). It therefore has the transformation property

$$\Phi_n(x, \theta) \longrightarrow \sum_m \left[\exp \left(i \sum_A t_A \Lambda^A(x_+) \right) \right]_{nm} \Phi_m(x, \theta). \quad (27.1.2)$$

If a term in the action depends only on left-chiral superfields, and not their derivatives or complex conjugates, like the term $\int d^4x [f(\Phi)]_{\mathcal{F}}$ in Eq. (??), then it (and its complex conjugate) will be invariant under the local transformation (27.1.2) if it is invariant under global transformations with $\Lambda^A(x)$ independent of x^μ . The need for introducing gauge fields in renormalizable theories of chiral superfields arises only in the D -terms, which involve both Φ_n and Φ_n^* . Because the matrices t_A are Hermitian, the Hermitian adjoint of Eq. (27.1.2) is

$$\Phi_n^\dagger(x, \theta) \longrightarrow \sum_m \Phi_m^\dagger(x, \theta) \left[\exp \left(-i \sum_A t_A \Lambda^A(x_+)^* \right) \right]_{mn}. \quad (27.1.3)$$

If it were not for the difference between $\Lambda^A(x_+)^* = \Lambda^A(x_-)$ and $\Lambda^A(x_+)$, this would just say that $\Phi^\dagger$ transforms according to the representation of the gauge group that is contragredient to the representation furnished by Φ , and any function of Φ and $\Phi^\dagger$ that is invariant under global gauge transformations would also be invariant under local gauge transformations. Because x_+ and x_- are different, we must introduce a gauge connection matrix $\Gamma_{nm}(x, \theta)$, with the transformation property

$$\Gamma(x, \theta) \longrightarrow \exp \left(+i \sum_A t_A \Lambda^A(x_+)^* \right) \Gamma(x, \theta) \exp \left(-i \sum_A t_A \Lambda^A(x_+) \right). \quad (27.1.4)$$

Then by multiplying $\Phi^\dagger$ on the right with Γ we obtain a superfield that transforms as

$$\left[\Phi^\dagger(x, \theta) \Gamma(x, \theta) \right]_n \longrightarrow \sum_m \left[\Phi^\dagger(x, \theta) \Gamma(x, \theta) \right]_m \left[\exp \left(-i \sum_A t_A \Lambda^A(x_+) \right) \right]_{mn}, \quad (27.1.5)$$

so that any globally gauge-invariant function constructed from Φ and $\Phi^\dagger \Gamma$ (and not their derivatives or complex conjugates) will also be locally gauge-invariant. One obvious example is the gauge-invariant version $(\Phi^\dagger \Gamma \Phi)_D$ of the D -term in the Lagrangian constructed in Section 26.4.

Any $\Gamma(x, \theta)$ that transforms as in Eq. (27.1.4) will allow us to construct gauge-invariant Lagrangians of chiral superfields. The choice is not unique; if Γ transforms as in Eq. (27.1.4), and we multiply on the right with any left-chiral superfield Υ_L with the transformation rule

$$\Upsilon_L(x, \theta) \longrightarrow \exp \left(i \sum_A t_A \Lambda^A(x_+) \right) \Upsilon_L(x, \theta) \exp \left(-i \sum_A t_A \Lambda^A(x_+) \right),$$

then we obtain a new gauge connection that also satisfies Eq. (27.1.4). One simplification is to take $\Gamma(x, \theta)$ Hermitian:

$$\Gamma^\dagger(x, \theta) = \Gamma(x, \theta). \quad (27.1.6)$$

This is always possible if there is any $\Gamma(x, \theta)$ that satisfies Eq. (27.1.4), for then by taking the Hermitian adjoint of Eq. (27.1.4) we easily see that $\Gamma^\dagger(x, \theta)$ transforms in the same way as $\Gamma(x, \theta)$, so if $\Gamma(x, \theta)$ is not Hermitian then we can replace it with its Hermitian part $(\Gamma + \Gamma^\dagger)/2$ (or, if that vanishes, by its anti-Hermitian part $(\Gamma - \Gamma^\dagger)/2i$.) Another simplification of great physical importance is to express $\Gamma(x, \theta)$ in terms of fields whose gauge transformation properties do not depend on the specific representation t_A of the gauge algebra under which the chiral superfield $\Phi(x, \theta)$ transforms, so that these fields can be used to form a suitable matrix $\Gamma(x, \theta)$ for chiral superfields that transform according to any representation of the gauge group. For this purpose, it is useful to recall the Baker–Hausdorff formula, which states that, for arbitrary matrices a and b ,

$$e^a e^b = \exp \left(a + b + \frac{1}{2}[a, b] + \frac{1}{12}[a, [a, b]] + \frac{1}{12}[b, [b, a]] + \dots \right), \quad (27.1.7)$$

where ‘ $\dots$ ’ denotes higher-order terms that can be written as multiple commutators of a s and b s, like the second- and third-order terms that are shown explicitly. It follows from this that for any representation of a Lie algebra, we have

$$\exp \left(\sum_A a^A t_A \right) \exp \left(\sum_A b^A t_A \right) = \exp \left(\sum_A f^A(a, b) t_A \right), \quad (27.1.8)$$

where

$$\begin{aligned} f^A(a, b) = & a^A + b^A + \frac{i}{2} \sum_{BC} C^A_{BC} a^B b^C - \frac{1}{12} \sum_{BCDE} C^A_{BC} C^C_{DE} a^B a^D b^E \\ & - \frac{1}{12} \sum_{BCDE} C^A_{BC} C^C_{DE} b^B b^D a^E + \dots, \end{aligned} \quad (27.1.9)$$

which depends on the Lie algebra through its structure constants C^A_{BC} , defined as usual by

$$[t_B, t_C] = i \sum_A C^A_{BC} t_A,$$

but does not depend on the particular representation furnished by the t_A . We will therefore take $\Gamma(x, \theta)$ in the form

$$\Gamma(x, \theta) = \exp \left(-2 \sum_A t_A V^A(x, \theta) \right), \quad (27.1.10)$$

where $V^A(x, \theta)$ are a set of real superfields (so that Γ is Hermitian), not depending on the representation of the gauge algebra furnished by the t_A .

We can achieve an important further simplification by noting an additional symmetry of supersymmetric gauge theories. If some function of Φ and $\Phi^\dagger \Gamma$ is invariant under global

gauge transformations, then it will automatically be invariant not only under the local gauge transformations (27.1.2)–(27.1.4), but also under the larger group of *extended gauge transformations*

$$\Phi_{nL}(x, \theta) \longrightarrow \sum_m \left[\exp \left(i \sum_A t_A \Omega^A(x, \theta) \right) \right]_{nm} \Phi_{mL}(x, \theta) \quad (27.1.11)$$

and

$$\Gamma(x, \theta) \longrightarrow \exp \left(-i \sum_A t_A \Omega^A(x, \theta) \right) \Gamma(x, \theta) \exp \left(+i \sum_A t_A \Omega^A(x, \theta)^* \right), \quad (27.1.12)$$

where $\Omega^A(x, \theta)$ is an arbitrary left-chiral superfield—that is, an arbitrary function of θ_L as well as x_+ . Under this transformation,

$$V^A(x, \theta) \longrightarrow V^A(x, \theta) + \frac{i}{2} [\Omega^A(x, \theta) - \Omega^A(x, \theta)^*] + \dots, \quad (27.1.13)$$

where ‘ $\dots$ ’ denotes terms arising from the commutators in Eq. (27.1.7), which are of first or higher order in gauge coupling constants. As a general left-chiral superfield, Ω may be written in the form (??)

$$\begin{aligned} \Omega^A(x, \theta) = & W^A(x) - \sqrt{2} \left(\bar{\theta} \frac{1 + \gamma_5}{2} w^A(x) \right) + \mathcal{W}^A(x) \left(\bar{\theta} \frac{1 + \gamma_5}{2} \theta \right) \\ & + \frac{1}{2} (\bar{\theta} \gamma_5 \gamma_\mu \theta) \partial^\mu W^A(x) - \frac{1}{\sqrt{2}} (\bar{\theta} \gamma_5 \theta) \left(\bar{\theta} \not{\partial} \frac{1 + \gamma_5}{2} w^A(x) \right) \\ & - \frac{1}{8} (\bar{\theta} \gamma_5 \theta)^2 \square W^A(x), \end{aligned} \quad (27.1.14)$$

in which $W^A(x)$ and $\mathcal{W}^A(x)$ are arbitrary complex functions of x^μ , and we introduced Majorana spinors $w^A(x)$ defined so that the left-handed spinor components of the superfields are $\frac{1}{2}(1 + \gamma_5)w^A(x)$. Using the complex conjugation properties (??) of Majorana bilinears, the complex conjugate of Eq. (27.1.14) gives

$$\begin{aligned} \Omega^A(x, \theta)^* = & W^{A*}(x) - \sqrt{2} \left(\bar{\theta} \frac{1 - \gamma_5}{2} w^A(x) \right) + \mathcal{W}^{A*}(x) \left(\bar{\theta} \frac{1 - \gamma_5}{2} \theta \right) \\ & - \frac{1}{2} (\bar{\theta} \gamma_5 \gamma_\mu \theta) \partial^\mu W^{A*}(x) - \frac{1}{\sqrt{2}} (\bar{\theta} \gamma_5 \theta) \left(\bar{\theta} \not{\partial} \frac{1 - \gamma_5}{2} w^A(x) \right) \\ & - \frac{1}{8} (\bar{\theta} \gamma_5 \theta)^2 \square W^{A*}(x). \end{aligned} \quad (27.1.15)$$

We write the real superfields $V^A(x, \theta)$ in terms of component fields as in Eq. (??):

$$\begin{aligned} V^A(x, \theta) = & C^A(x) - i(\bar{\theta} \gamma_5 \omega^A(x)) - \frac{i}{2} (\bar{\theta} \gamma_5 \theta) M^A(x) - \frac{1}{2} (\bar{\theta} \theta) N^A(x) \\ & + \frac{i}{2} (\bar{\theta} \gamma_5 \gamma^\mu \theta) V_\mu^A(x) - i(\bar{\theta} \gamma_5 \theta) \left(\bar{\theta} \left[\lambda^A(x) + \frac{1}{2} \not{\partial} \omega^A(x) \right] \right) \\ & - \frac{1}{4} (\bar{\theta} \gamma_5 \theta)^2 \left(D^A(x) + \frac{1}{2} \square C^A(x) \right), \end{aligned} \quad (27.1.16)$$

where $C^A(x)$, $M^A(x)$, $N^A(x)$, and $V_\mu^A(x)$ are all real, and $\omega^A(x)$ and $\lambda^A(x)$ are Majorana spinors. Using Eqs. (27.1.14)–(27.1.16) in Eq. (27.1.13), we find that the component fields of the gauge superfield undergo the extended gauge transformation

$$\begin{aligned}
C^A(x) &\longrightarrow C^A(x) - \text{Im } W^A(x) + \cdots, \\
\omega^A(x) &\longrightarrow \omega^A(x) + \frac{1}{\sqrt{2}} w^A(x) + \cdots, \\
V_\mu^A(x) &\longrightarrow V_\mu^A(x) + \partial_\mu \text{Re } W^A(x) + \cdots, \\
M^A(x) &\longrightarrow M^A(x) - \text{Re } \mathcal{W}^A(x) + \cdots, \\
N^A(x) &\longrightarrow N^A(x) + \text{Im } \mathcal{W}^A(x) + \cdots, \\
\lambda^A(x) &\longrightarrow \lambda^A(x) + \cdots, \\
D^A(x) &\longrightarrow D^A(x) + \cdots,
\end{aligned} \tag{27.1.17}$$

where again ‘ $\cdots$ ’ denotes terms that arise from the structure constants in Eq. (27.1.9) and that are therefore proportional to one or more factors of gauge coupling constants. We can use such an extended gauge transformation to put the gauge superfields into a convenient form, known as *Wess–Zumino gauge* [1], in which

$$C^A(x) = \omega^A(x) = M^A(x) = N^A(x) = 0, \tag{27.1.18}$$

so that

$$V^A(x, \theta) = \frac{i}{2} (\bar{\theta} \gamma_5 \gamma^\mu \theta) V_\mu^A(x) - i (\bar{\theta} \gamma_5 \theta) (\bar{\theta} \lambda^A(x)) - \frac{1}{4} (\bar{\theta} \gamma_5 \theta)^2 D^A(x). \tag{27.1.19}$$

To accomplish this to zeroth order in the coupling constants, it is only necessary to set $\text{Im } W^A(x) = C^A(x)$, $w^A(x) = -\sqrt{2} \omega^A(x)$, and $\mathcal{W}^A(x) = M^A(x) - i N^A(x)$. For Abelian gauge theories, in which the structure constants vanish, this ends our task. For non-Abelian gauge theories it is necessary to add terms to $\text{Im } W^A(x)$, $w^A(x)$, and $\mathcal{W}^A(x)$ of first order in gauge coupling constants to cancel the terms arising from commutators of zeroth-order terms, then add terms to $\text{Im } W^A(x)$, $w^A(x)$, and $\mathcal{W}^A(x)$ of second order in gauge coupling constants to cancel the terms arising from commutators of first-order terms with zeroth-order terms, and so on. It is not easy to calculate the series of terms in $\text{Im } W^A(x)$, $w^A(x)$, and $\mathcal{W}^A(x)$ needed to satisfy the gauge conditions (27.1.18) to all orders in gauge couplings, but there is no need—the important thing is that it is possible.

Inspection of the transformation rules (??)–(??) shows that the Wess–Zumino gauge condition (27.1.18) is not invariant under supersymmetry transformations unless $V_\mu^A = \lambda^A = 0$, and the condition $\lambda^A = 0$ is not supersymmetric unless also $D^A = 0$, in which case the whole superfield vanishes. Once we adopt Wess–Zumino gauge, the action is no longer invariant under either general extended gauge transformations or under supersymmetry, but it is invariant under supersymmetry transformations, which take us out of Wess–Zumino gauge, followed by suitable extended gauge transformations that take us back to Wess–Zumino gauge. (We will go into this explicitly in Section 27.8.) As we shall now see, it is also invariant under the ordinary gauge transformations (27.1.2)–(27.1.4), which preserve Wess–Zumino gauge.

With the gauge superfield satisfying the Wess–Zumino gauge condition (27.1.18), it becomes relatively easy to calculate its behavior under ordinary infinitesimal gauge transformations. In this case, $\Omega^A(x_+)$ are left-chiral superfields of the form (??), but with no ψ_L - or $\mathcal{F}$ -components, and with ϕ -components given by *real* infinitesimal functions $\Lambda^A(x)$:

$$\Omega^A(x_+) = \Lambda^A(x) + \frac{1}{2}(\bar{\theta}\gamma_5\gamma_\mu\theta)\partial^\mu\Lambda^A(x) - \frac{1}{8}(\bar{\theta}\gamma_5\theta)^2\Box\Lambda^A(x). \quad (27.1.20)$$

To calculate the product of exponentials in the transformation rule (27.1.4), we use a version of the Baker–Hausdorff formula:

$$\exp(a)\exp(X)\exp(b) = \exp[X + L_X \cdot (b - a) + (L_X \coth L_X) \cdot (b + a) + \cdots], \quad (27.1.21)$$

where a , b , and X are arbitrary matrices, L_X is the operator

$$L_X \cdot f = \frac{1}{2}[X, f], \quad (27.1.22)$$

and ‘ $\cdots$ ’ here denotes terms of second and higher order in a and/or b . In our case we have

$$\begin{aligned} b + a &= 2 \sum_A t_A \operatorname{Im} \Lambda^A(x_+) = -i(\bar{\theta}\gamma_5\gamma_\mu\theta) \sum_A t_A \partial^\mu \Lambda^A(x), \\ b - a &= -2i \sum_A t_A \operatorname{Re} \Lambda^A(x_+) = -2i \sum_A t_A \left[\Lambda^A(x) - \frac{1}{8}(\bar{\theta}\gamma_5\theta)^2\Box\Lambda^A(x) \right], \\ X &= -2 \sum_A t_A V^A(x, \theta) \\ &= -2 \sum_A t_A \left[\frac{i}{2}(\bar{\theta}\gamma_5\gamma^\mu\theta)V_\mu^A(x) - i(\bar{\theta}\gamma_5\theta)(\bar{\theta}\lambda^A(x)) - \frac{1}{4}(\bar{\theta}\gamma_5\theta)^2 D^A(x) \right]. \end{aligned}$$

Now, every term in X involves at least one factor of θ_L and at least one factor of θ_R , while $a + b$ has just one factor of θ_L and one factor of θ_R , so we can drop any terms in $L_X \coth L_X$ of second or higher order in L_X . Since $L_X \coth L_X$ is an *even* function of L_X , this means that we can replace it with its term of zeroth order in L_X , which is just unity. Also, we may drop the term in $b - a$ proportional to $(\bar{\theta}\gamma_5\theta)^2$, since when acted on by L_X it would yield at least three factors of either θ_L or θ_R . Thus the argument of the exponential on the right-hand side of Eq. (27.1.21) may be replaced with

$$\begin{aligned} X + \frac{1}{2}[X, b - a] + b + a &= -2 \sum_A t_A \left[V^A(x, \theta) \right. \\ &\quad + \sum_{BC} C^A_{BC} V^B(x, \theta) \Lambda^C(x) \\ &\quad \left. + \frac{i}{2}(\bar{\theta}\gamma_5\gamma_\mu\theta)\partial^\mu\Lambda^A(x) \right]. \end{aligned}$$

Thus for infinitesimal gauge transformations, the transformation rule (27.1.4) yields

$$V^A(x, \theta) \longrightarrow V^A(x, \theta) + \sum_{BC} C^A_{BC} V^B(x, \theta) \Lambda^C(x) + \frac{i}{2}(\bar{\theta}\gamma_5\gamma_\mu\theta)\partial^\mu\Lambda^A(x). \quad (27.1.23)$$

It is important to note that under ordinary gauge transformations, a gauge superfield in Wess–Zumino gauge remains in Wess–Zumino gauge. In terms of the component fields in Eq. (27.1.19), Eq. (27.1.23) reads

$$V_\mu^A(x) \longrightarrow \sum_{BC} C^A{}_{BC} V_\mu^B(x) \Lambda^C(x) + \partial_\mu \Lambda^A(x), \quad (27.1.24)$$

$$\lambda^A(x) \longrightarrow \sum_{BC} C^A{}_{BC} \lambda^B(x) \Lambda^C(x), \quad (27.1.25)$$

$$D^A(x) \longrightarrow \sum_{BC} C^A{}_{BC} D^B(x) \Lambda^C(x). \quad (27.1.26)$$

We recognize Eq. (27.1.24) as the usual Yang–Mills gauge transformation rule (15.1.9) of a gauge field, while Eqs. (27.1.25) and (27.1.26) tell us that the fields $\lambda^A(x)$ and $D^A(x)$ transform as ‘matter’ fields that belong to the adjoint representation of the gauge group. The Majorana spinors λ^A are known as *gaugino* fields, while the real scalars D^A will turn out to be another set of auxiliary fields.

Next we must evaluate the matrix Γ that is needed in constructing gauge-invariant functions of chiral superfields. Because all terms with more than four factors of θ vanish, the expansion of the exponential is very simple in Wess–Zumino gauge:

$$\begin{aligned} \Gamma(x, \theta) &= \exp \left(-2 \sum_A t_A V^A(x, \theta) \right) \\ &= \mathbf{1} - i(\bar{\theta} \gamma_5 \gamma^\mu \theta) \sum_A t_A V_\mu^A(x) \\ &\quad - \frac{1}{2}(\bar{\theta} \gamma_5 \gamma^\mu \theta)(\bar{\theta} \gamma_5 \gamma^\nu \theta) \sum_{AB} t_A t_B V_\mu^A(x) V_\nu^B(x) \\ &\quad + 2i(\bar{\theta} \gamma_5 \theta) \sum_A t_A (\bar{\theta} \lambda^A(x)) + \frac{1}{2}(\bar{\theta} \gamma_5 \theta)^2 \sum_A t_A D^A(x). \end{aligned}$$

We can construct a gauge-invariant density by multiplying this on the right with a column vector of left-chiral superfields of the form (??):

$$\begin{aligned} \Phi_n(x, \theta) &= \phi_n(x) - \sqrt{2}(\bar{\theta} \psi_{nL}(x)) + \mathcal{F}_n(x) \left(\bar{\theta} \frac{1 + \gamma_5}{2} \theta \right) \\ &\quad + \frac{1}{2}(\bar{\theta} \gamma_5 \gamma_\mu \theta) \partial^\mu \phi_n(x) - \frac{1}{\sqrt{2}}(\bar{\theta} \gamma_5 \theta)(\bar{\theta} \not{\partial} \psi_{nL}(x)) - \frac{1}{8}(\bar{\theta} \gamma_5 \theta)^2 \square \phi_n(x), \end{aligned}$$

and multiplying on the left with the column

$$\begin{aligned} \Phi_n(x, \theta)^* &= \phi_n^*(x) - \sqrt{2}(\overline{\psi_{nL}(x)} \theta) + \mathcal{F}_n^*(x) \left(\bar{\theta} \frac{1 - \gamma_5}{2} \theta \right) \\ &\quad - \frac{1}{2}(\bar{\theta} \gamma_5 \gamma_\mu \theta) \partial^\mu \phi_n^*(x) - \frac{1}{\sqrt{2}}(\bar{\theta} \gamma_5 \theta) \partial_\mu (\overline{\psi_{nL}(x)} \gamma^\mu \theta) \\ &\quad - \frac{1}{8}(\bar{\theta} \gamma_5 \theta)^2 \square \phi_n^*(x). \end{aligned}$$

The term in this product of fourth order in θ is

$$\begin{aligned}
\left[\Phi^\dagger \Gamma \Phi\right]_{\theta^4} = & -\frac{1}{8}(\bar{\theta}\gamma_5\theta)^2 \left\{ [\phi^\dagger \square \phi] + [(\square \phi^\dagger)\phi] \right\} \\
& + (\bar{\theta}\gamma_5\theta) \left\{ [(\bar{\psi}_L\theta)(\bar{\theta}\gamma^\mu\partial_\mu\psi_L)] + [((\partial_\mu\bar{\psi}_L)\gamma^\mu\theta)(\bar{\theta}\psi_L)] \right\} \\
& + \frac{1}{4}(\bar{\theta}(1-\gamma_5)\theta)(\bar{\theta}(1+\gamma_5)\theta)[\mathcal{F}^\dagger\mathcal{F}] \\
& - \frac{1}{4}(\bar{\theta}\gamma_5\gamma^\mu\theta)(\bar{\theta}\gamma_5\gamma^\nu\theta)[\partial_\mu\phi^\dagger\partial_\nu\phi] \\
& - \frac{i}{2}(\bar{\theta}\gamma_5\gamma^\mu\theta)(\bar{\theta}\gamma_5\gamma^\nu\theta)\sum_A V_\mu^A \left\{ [\phi^\dagger t_A\partial_\nu\phi] - [(\partial_\nu\phi^\dagger)t_A\phi] \right\} \\
& - \frac{1}{2}(\bar{\theta}\gamma_5\gamma^\mu\theta)(\bar{\theta}\gamma_5\gamma^\nu\theta)\sum_{AB} V_\mu^A V_\nu^B [\phi^\dagger t_A t_B \phi] \\
& - 2i(\bar{\theta}\gamma_5\gamma_\mu\theta)\sum_A V^{A\mu}[(\bar{\psi}_L\theta)t_A(\bar{\theta}\psi_L)] \\
& - 2i\sqrt{2}(\bar{\theta}\gamma_5\theta)\sum_A [(\bar{\psi}_L\theta)t_A(\bar{\theta}\lambda^A)\phi] \\
& - 2i\sqrt{2}(\bar{\theta}\gamma_5\theta)\sum_A [\phi^\dagger(\bar{\lambda}^A\theta)t_A(\bar{\theta}\psi_L)] \\
& + \frac{1}{2}(\bar{\theta}\gamma_5\theta)^2\sum_A D_A[\phi^\dagger t_A\phi],
\end{aligned}$$

in which we use square brackets to indicate scalar products in the flavor indices n, m , and continue to use round brackets to indicate scalar products in Dirac indices. Just as in Section 26.4, we may use the identities (??)–(??) to put all θ dependence in the form of an over-all factor $(\bar{\theta}\gamma_5\theta)^2$:

$$\begin{aligned}
\left[\Phi^\dagger \Gamma \Phi\right]_{\theta^4} = & (\bar{\theta}\gamma_5\theta)^2 \left\{ -\frac{1}{8}[\phi^\dagger \square \phi] - \frac{1}{8}[(\square \phi^\dagger)\phi] \right. \\
& + \frac{1}{4}[(\bar{\psi}_L\gamma^\mu\partial_\mu\psi_L)] - \frac{1}{4}[(\partial_\mu\bar{\psi}_L)\gamma^\mu\psi_L] \\
& - \frac{1}{2}[\mathcal{F}^\dagger\mathcal{F}] + \frac{1}{4}[\partial_\mu\phi^\dagger\partial^\mu\phi] \\
& + \frac{i}{2}\sum_A V_\mu^A[\phi^\dagger t_A\partial^\mu\phi] - \frac{i}{2}\sum_A V_\mu^A[(\partial^\mu\phi^\dagger)t_A\phi] \\
& + \frac{1}{2}\sum_{AB} V_\mu^A V^{B\mu}[\phi^\dagger t_A t_B \phi] - \frac{i}{2}\sum_A V_\mu^A[(\bar{\psi}_L\gamma^\mu t_A\psi_L)] \\
& - \frac{i}{\sqrt{2}}\sum_A [(\bar{\psi}_L t_A \lambda^A)\phi] + \frac{i}{\sqrt{2}}\sum_A [\phi^\dagger(\bar{\lambda}^A t_A \psi_L)] \\
& \left. + \frac{1}{2}\sum_A D_A[\phi^\dagger t_A\phi] \right\}.
\end{aligned}$$

The D -term is the coefficient of $-\frac{1}{4}(\bar{\theta}\gamma_5\theta)^2$ minus $\frac{1}{2}\square$ acting on the θ -independent term,

which for $[\Phi^\dagger \Gamma \Phi]$ is $[\phi^\dagger \phi]$, so

$$\begin{aligned}
[\Phi^\dagger \Gamma \Phi]_D = & -2[\partial_\mu \phi^\dagger \partial^\mu \phi] - [(\overline{\psi}_L \gamma^\mu \partial_\mu \psi_L)] + [(\partial_\mu \overline{\psi}_L) \gamma^\mu \psi_L] + 2[\mathcal{F}^\dagger \mathcal{F}] \\
& - 2i \sum_A V_\mu^A [\phi^\dagger t_A \partial^\mu \phi] + 2i \sum_A V_\mu^A [(\partial^\mu \phi^\dagger) t_A \phi] \\
& - 2 \sum_{AB} V_\mu^A V^{B\mu} [\phi^\dagger t_A t_B \phi] + 2i \sum_A V_\mu^A [(\overline{\psi}_L \gamma^\mu t_A \psi_L)] \\
& + 2i\sqrt{2} \sum_A [(\overline{\psi}_L t_A \lambda^A) \phi] - 2i\sqrt{2} \sum_A [\phi^\dagger (\overline{\lambda^A} t_A \psi_L)] \\
& - 2 \sum_A D_A [\phi^\dagger t_A \phi].
\end{aligned}$$

To see that this is gauge-invariant, we note that it may be written as

$$\begin{aligned}
\frac{1}{2}[\Phi^\dagger \Gamma \Phi]_D = & -[(D_\mu \phi)^\dagger D^\mu \phi] - \frac{1}{2}[(\overline{\psi}_L \gamma^\mu D_\mu \psi_L)] + \frac{1}{2}[(\overline{D_\mu \psi_L} \gamma^\mu \psi_L)] + [\mathcal{F}^\dagger \mathcal{F}] \\
& + i\sqrt{2} \sum_A [(\overline{\psi}_L t_A \lambda^A) \phi] - i\sqrt{2} \sum_A [\phi^\dagger (\overline{\lambda^A} t_A \psi_L)] - \sum_A D_A [\phi^\dagger t_A \phi],
\end{aligned} \tag{27.1.27}$$

where D_μ is the gauge-invariant derivative (15.1.10):

$$D_\mu \psi_L \equiv \partial_\mu \psi_L - i \sum_A t_A V_\mu^A \psi_L, \quad D_\mu \phi \equiv \partial_\mu \phi - i \sum_A t_A V_\mu^A \phi. \tag{27.1.28}$$

Eq. (27.1.27) is thus a suitable gauge-invariant kinematic Lagrangian for the scalar and spinor components of a left-chiral superfield, now supplemented with Yukawa couplings of gaugino fields to the scalar and spinor components of chiral superfields as well as terms involving auxiliary fields $\mathcal{F}_n$ and D_A .

27.2 Gauge-Invariant Action for Abelian Gauge Superfields

We must now consider how to construct a gauge-invariant supersymmetric action for the gauge superfields $V^A(x, \theta)$ that contain the gauge fields $V_\mu^A(x)$. In order to motivate this construction we will first consider the case of a single Abelian gauge field (dropping the superscript A), and then return to the general case in the next section.

In an Abelian gauge theory like quantum electrodynamics, the gauge-invariant field constructed from $V_\mu(x)$ is the familiar field-strength tensor

$$f_{\mu\nu}(x) = \partial_\mu V_\nu(x) - \partial_\nu V_\mu(x). \tag{27.2.1}$$

The supersymmetry transformation rule for $f_{\mu\nu}(x)$ is then given by the transformation rule (??) for $V_\mu(x)$ as

$$\delta f_{\mu\nu} = (\bar{\alpha}(\partial_\mu \gamma_\nu - \partial_\nu \gamma_\mu) \lambda). \tag{27.2.2}$$

Eq. (??) gives the transformation rule for $\lambda(x)$ as

$$\delta \lambda = \left(-\frac{1}{4} f_{\mu\nu} [\gamma^\mu, \gamma^\nu] + i\gamma_5 D \right) \alpha, \tag{27.2.3}$$

while Eq. (??) gives the transformation rule for $D(x)$:

$$\delta D = i(\bar{\alpha}\gamma_5\not{\partial}\lambda). \quad (27.2.4)$$

None of this depends on whether or not the superfield $V^A(x, \theta)$ is taken to be in Wess–Zumino gauge. We see that the fields $f_{\mu\nu}(x)$, $\lambda(x)$, and $D(x)$ form a complete supersymmetry multiplet.

It is not hard to construct a suitable kinematic Lagrangian density for the fields of this supermultiplet. The only Lorentz-invariant, parity conserving, and gauge-invariant functions of these fields with dimensionality four are $f_{\mu\nu}f^{\mu\nu}$, $\bar{\lambda}\not{\partial}\lambda$, and D^2 . We can make V^μ a conventionally normalized vector field by taking the coefficient of $f_{\mu\nu}f^{\mu\nu}$ to be $-1/4$, so we may tentatively take the kinematic Lagrangian density as

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}f_{\mu\nu}f^{\mu\nu} - c_\lambda(\bar{\lambda}\not{\partial}\lambda) - c_D D^2,$$

with coefficients c_λ and c_D to be determined from the condition that $\int \mathcal{L}_{\text{gauge}} d^4x$ is supersymmetric. Using Eqs. (27.2.2)–(27.2.4), an infinitesimal supersymmetry transformation changes the operators in the Lagrangian by

$$\begin{aligned} \delta(f_{\mu\nu}f^{\mu\nu}) &= 2f^{\mu\nu}(\bar{\alpha}(\gamma_\nu\partial_\mu - \gamma_\mu\partial_\nu)\lambda), \\ \delta(\bar{\lambda}\not{\partial}\lambda) &= 2\left(\bar{\alpha}\left[-\frac{1}{4}f_{\mu\nu}[\gamma^\mu, \gamma^\nu] + i\gamma_5 D\right]\not{\partial}\lambda\right), \\ \delta D^2 &= 2iD(\bar{\alpha}\gamma_5\not{\partial}\lambda), \end{aligned}$$

in which we drop derivative terms that do not contribute to the variation of the action. In order to see how these terms cancel, it is necessary to use an identity for gamma matrices¹

$$[\gamma^\mu, \gamma^\nu]\gamma^\rho = -2\eta^{\mu\rho}\gamma^\nu + 2\eta^{\nu\rho}\gamma^\mu - 2i\epsilon^{\mu\nu\rho\sigma}\gamma_\sigma\gamma_5. \quad (27.2.5)$$

The term $-i\epsilon^{\mu\nu\rho\sigma}f_{\mu\nu}(\bar{\alpha}\gamma_\sigma\gamma_5\partial_\rho\lambda)$ does not contribute to $\int d^4x \delta\mathcal{L}$, because integrating by parts yields a contribution proportional to $\epsilon^{\mu\nu\rho\sigma}\partial_\rho f_{\mu\nu}$, which vanishes for an $f_{\mu\nu}$ of the form (27.2.1). This identity then allows us to rewrite the variation of the λ -term as

$$\delta(\bar{\lambda}\not{\partial}\lambda) = -f^{\mu\nu}(\bar{\alpha}(\gamma_\nu\partial_\mu - \gamma_\mu\partial_\nu)\lambda) + 2iD(\bar{\alpha}\gamma_5\not{\partial}\lambda).$$

The cancellation of terms proportional to $f^{\mu\nu}\lambda$ requires that $c_\lambda = 1/2$, while the cancellation of terms proportional to $D\lambda$ requires that $c_D = -c_\lambda$, so the supersymmetric Lagrangian density takes the form

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}f_{\mu\nu}f^{\mu\nu} - \frac{1}{2}(\bar{\lambda}\not{\partial}\lambda) + \frac{1}{2}D^2. \quad (27.2.6)$$

This shows that with V^μ canonically normalized, the field λ related to V^μ by the transformation rules (27.2.2) and (27.2.3) is also canonically normalized.

¹To derive this, use the fact that any $4 \cdot 4$ matrix may be expressed as a linear combination of the 16 independent covariant matrices described in Section 5.4, which in our case is limited by Lorentz invariance and space inversion invariance to the terms shown here. The coefficients of these terms can be calculated by giving $\mu\nu\rho$ the values 121 and 123.

In addition, for Abelian gauge theories there is a superrenormalizable term, known as a *Fayet–Iliopoulos term* [2]:

$$\mathcal{L}_{\text{FI}} = \xi D, \quad (27.2.7)$$

with ξ an arbitrary constant. Its variation under a supersymmetry transformation is shown by Eq. (27.2.4) to be a derivative, so that it yields another supersymmetric term in the action. As we will see in Section 27.5, the presence of such a term can provide a mechanism for the spontaneous breakdown of supersymmetry.

Both for its own interest and as a tool in constructing supersymmetric interactions involving the fields $f_{\mu\nu}$, λ , and D , it is interesting to ask what sort of superfield has these as component fields. Somewhat surprisingly, it turns out to be a *spinor superfield* $W_\alpha(x)$, with component fields (in the notation of Eq. (??)) given by

$$\begin{aligned} C_{(\alpha)}(x) &= \lambda_\alpha(x), \\ \omega_{(\alpha)\beta}(x) &= \frac{i}{2}(\gamma^\mu \gamma^\nu \epsilon)_{\alpha\beta} f_{\mu\nu}(x) + (\gamma_5 \epsilon)_{\alpha\beta} D(x), \\ V_{(\alpha)\mu}(x) &= -i\partial_\mu(\gamma_5 \lambda(x))_\alpha, \\ M_{(\alpha)}(x) &= -i(\not{\partial} \gamma_5 \lambda(x))_\alpha, \quad N_{(\alpha)}(x) = -(\not{\partial} \lambda(x))_\alpha, \\ \lambda_{(\alpha)\beta}(x) &= D_{(\alpha)}(x) = 0. \end{aligned} \quad (27.2.8)$$

(The subscript α on these component fields is put inside parentheses to emphasize that it labels the whole superfield.) It is straightforward to use Eqs. (27.2.2)–(27.2.4) to check directly that the superfield components given by Eq. (27.2.8) do transform as in Eqs. (??)–(??).

Inserting the component fields (27.2.8) in Eq. (??), and using Eq. (??), we find that the superfield W_α takes the form

$$\begin{aligned} W_\alpha(x, \theta) &= \left[\lambda_\alpha(x) + \frac{1}{2} \gamma^\mu \gamma^\nu \theta f_{\mu\nu}(x) - i \gamma_5 \theta D(x) - \frac{1}{2} (\theta^T \epsilon \theta) \not{\partial} \gamma_5 \lambda_\alpha(x) \right. \\ &\quad + \frac{1}{2} (\theta^T \epsilon \gamma_5 \theta) \not{\partial} \lambda_\alpha(x) + \frac{1}{2} (\theta^T \epsilon \gamma^\mu \theta) \gamma_5 \partial_\mu \lambda_\alpha(x) \\ &\quad - \frac{1}{4} (\theta^T \epsilon \theta) \gamma_5 \gamma^\mu \gamma^\nu \gamma^\sigma \theta \partial_\sigma f_{\mu\nu}(x) \\ &\quad \left. + \frac{i}{2} (\theta^T \epsilon \theta) \gamma^\sigma \theta \partial_\sigma D(x) - \frac{1}{8} (\theta^T \epsilon \theta)^2 \square \lambda_\alpha(x) \right]. \end{aligned} \quad (27.2.9)$$

As we showed in Section 26.3, a superfield like this with zero λ - and D -components is *chiral*—that is, it is the sum of left-chiral and right-chiral superfields

$$W(x, \theta) = W_L(x, \theta) + W_R(x, \theta). \quad (27.2.10)$$

Here the left- and right-chiral superfields are simply the projections of W on the subspaces with $\gamma_5 = +1$ and $\gamma_5 = -1$, respectively:

$$\begin{aligned} W_L(x, \theta) &= \frac{1}{2}(1 + \gamma_5)W(x, \theta) \\ &= \lambda_L(x_+) + \frac{1}{2} \gamma^\mu \gamma^\nu \theta_L f_{\mu\nu}(x_+) + (\theta_L^T \epsilon \theta_L) \not{\partial} \lambda_R(x_+) - i \theta_L D(x_+). \end{aligned} \quad (27.2.11)$$

$$\begin{aligned}
W_R(x, \theta) &= \frac{1}{2}(1 - \gamma_5)W(x, \theta) \\
&= \lambda_R(x_-) + \frac{1}{2}\gamma^\mu\gamma^\nu\theta_R f_{\mu\nu}(x_-) - (\theta_R^T\epsilon\theta_R)\not\partial\lambda_L(x_-) - i\theta_R D(x_-),
\end{aligned} \tag{27.2.12}$$

with $x_\pm^\mu$ given by Eq. (??).

As we saw in Section 26.3, we can construct suitable Lagrangian densities from the $\mathcal{F}$ -term of any scalar function of a left-chiral superfield, plus its Hermitian adjoint. The simplest scalar function of the left-chiral superfield (27.2.11) is $\sum_{\alpha\beta}\epsilon_{\alpha\beta}W_{L\alpha}W_{L\beta}$. To calculate the $\mathcal{F}$ -term, we note that, when expressed as a function of θ_L and x_+ , the term in $\sum_{\alpha\beta}\epsilon_{\alpha\beta}W_{L\alpha}W_{L\beta}$ of second order in θ_L is

$$\begin{aligned}
-\left[\sum_{\alpha\beta}\epsilon_{\alpha\beta}W_{L\alpha}W_{L\beta}\right]_{\theta_L^2} &= (\theta_L^T\epsilon\theta_L)\left[-2(\lambda_L^T(x)\epsilon\not\partial\lambda_R(x)) + D^2(x)\right] \\
&\quad + \frac{1}{16}(\overline{\theta}_L[\gamma^\mu, \gamma^\nu][\gamma^\rho, \gamma^\sigma]\theta_L)f_{\mu\nu}(x)f_{\rho\sigma}(x).
\end{aligned}$$

(The argument of the fields here can be taken as x^μ rather than x_+^μ because the difference would yield terms that contain at least three factors of θ_L , which therefore vanish.) Lorentz invariance together with the fact that $(\bar{s}[\gamma_\mu, \gamma_\nu]s)$ and $(\bar{s}[\gamma_\mu, \gamma_\nu]\gamma_5 s)$ vanish for any Majorana fermion s tell us that the bilinear $(\overline{\theta}_L[\gamma^\mu, \gamma^\nu][\gamma^\rho, \gamma^\sigma]\theta_L)$ must be proportional to a linear combination of $(\overline{\theta}_L\theta_L)(\eta^{\mu\rho}\eta^{\nu\sigma} - \eta^{\mu\sigma}\eta^{\nu\rho})$ and $(\overline{\theta}_L\theta_L)\epsilon^{\mu\nu\rho\sigma}$. We can find the coefficients by giving $\mu\nu\rho\sigma$ the values 1212 or 1230, and in this way we find that

$$\begin{aligned}
(\overline{\theta}_L[\gamma^\mu, \gamma^\nu][\gamma^\rho, \gamma^\sigma]\theta_L) \\
= 4(\overline{\theta}_L\theta_L)\left[-\eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho} + i\epsilon^{\mu\nu\rho\sigma}\right].
\end{aligned}$$

The $\mathcal{F}$ -term is the coefficient of $(\overline{\theta}_L\theta_L)$, so

$$\begin{aligned}
-\left[\sum_{\alpha\beta}\epsilon_{\alpha\beta}W_{L\alpha}W_{L\beta}\right]_{\mathcal{F}} &= -2(\overline{\lambda}_R\not\partial\lambda_R) - \frac{1}{2}f_{\mu\nu}f^{\mu\nu} \\
&\quad + \frac{i}{4}\epsilon^{\mu\nu\rho\sigma}f_{\mu\nu}f_{\rho\sigma} + D^2.
\end{aligned} \tag{27.2.13}$$

Eq. (??) shows that $(\bar{\lambda}\not\partial\lambda)$ is real, while $(\bar{\lambda}\not\partial\gamma_5\lambda)$ is imaginary, so the real part of Eq. (27.2.13) yields the Lagrangian (27.2.6) for the gauge and gaugino fields

$$-\frac{1}{2}\text{Re}\left[\sum_{\alpha\beta}\epsilon_{\alpha\beta}W_{L\alpha}W_{L\beta}\right]_{\mathcal{F}} = -\frac{1}{2}(\bar{\lambda}\not\partial\lambda) - \frac{1}{4}f_{\mu\nu}f^{\mu\nu} + \frac{1}{2}D^2. \tag{27.2.14}$$

The physical significance of the imaginary part will be discussed in a more general context in the next section.

There is another way to derive the form of the spinor superfield, which will turn out to provide a more convenient way of deriving the components of the gauge superfield in non-Abelian gauge theories. A tedious but straightforward calculation shows that the gauge-invariant superfield (27.2.9) may be expressed in terms of the gauge superfield (27.1.16)

as

$$W_\alpha(x, \theta) = \frac{i}{4}(\mathcal{D}^T \epsilon \mathcal{D}) \mathcal{D}_\alpha V(x, \theta), \quad (27.2.15)$$

where $\mathcal{D}_\alpha$ is the superderivative introduced in Eq. (??):

$$\mathcal{D}_\alpha \equiv \sum_\beta (\gamma_5 \epsilon)_{\alpha\beta} \frac{\partial}{\partial \theta_\beta} - (\gamma^\mu \theta)_\alpha \frac{\partial}{\partial x^\mu} = -\frac{\partial}{\partial \bar{\theta}_\alpha} - (\gamma^\mu \theta)_\alpha \frac{\partial}{\partial x^\mu}.$$

This result might have been obtained (aside from the normalization factor) by noting that the function (27.2.15) has the desired property of being a gauge-invariant chiral spinor superfield. First, note that Eq. (27.2.15) is a superfield, because it is formed by acting on the superfield V with superderivatives. Also, it follows from the anticommutation of the $\mathcal{D}$ s that the product of any three or more $\mathcal{D}_L$ s or three or more $\mathcal{D}_R$ s vanishes, so that

$$(\mathcal{D}^T \epsilon \mathcal{D}) \mathcal{D} = (\mathcal{D}_L^T \epsilon \mathcal{D}_L) \mathcal{D}_R + (\mathcal{D}_R^T \epsilon \mathcal{D}_R) \mathcal{D}_L. \quad (27.2.16)$$

Because $\mathcal{D}_L(\mathcal{D}_L^T \epsilon \mathcal{D}_L) = \mathcal{D}_R(\mathcal{D}_R^T \epsilon \mathcal{D}_R) = 0$, the superfield (27.2.15) is chiral, with

$$\begin{aligned} W_{L\alpha}(x, \theta) &= \frac{i}{4}(\mathcal{D}_R^T \epsilon \mathcal{D}_R) \mathcal{D}_{L\alpha} V(x, \theta), \\ W_{R\alpha}(x, \theta) &= \frac{i}{4}(\mathcal{D}_L^T \epsilon \mathcal{D}_L) \mathcal{D}_{R\alpha} V(x, \theta). \end{aligned} \quad (27.2.17)$$

Finally, we can show that (27.2.15) is invariant under the generalized gauge transformation (27.1.13), which for a single Abelian gauge superfield is simply

$$V(x, \theta) \longrightarrow V(x, \theta) + \frac{i}{2} [\Omega(x, \theta) - \Omega^*(x, \theta)], \quad (27.2.18)$$

where $\Omega(x, \theta)$ is an arbitrary left-chiral superfield. Since $\mathcal{D}_L \Omega^* = 0$, the change in $W_{L\alpha}$ is proportional to $(\mathcal{D}_R^T \epsilon \mathcal{D}_R) \mathcal{D}_{L\alpha} \Omega$. But $\mathcal{D}_R \Omega = 0$ and

$$[(\mathcal{D}_R^T \epsilon \mathcal{D}_R), \mathcal{D}_{L\alpha}] = -2 [(1 + \gamma_5) \not{\mathcal{D}} \mathcal{D}_R]_\alpha,$$

so the change in $W_{L\alpha}$ vanishes. A similar argument shows that $W_{R\alpha}$ is also gauge-invariant. (The work of checking Eq. (27.2.15) is greatly reduced by using this gauge-invariance property to put $V(x, \theta)$ in Wess–Zumino gauge.)

The chiral superfields (27.2.11) and (27.2.12) are evidently not of the most general form for left- and right-chiral superfields. To put the constraints satisfied by these superfields in a manifestly supersymmetric form, we note by using the anticommutation relation (??) that

$$\begin{aligned} \epsilon_{\alpha\beta} \mathcal{D}_{L\alpha} (\mathcal{D}_R^T \epsilon \mathcal{D}_R) \mathcal{D}_{L\beta} &= -2 \mathcal{D}_{R\alpha} \mathcal{D}_{L\beta} (\epsilon (1 + \gamma_5) \not{\mathcal{D}})_{\beta\alpha} \\ &\quad + (\mathcal{D}_R^T \epsilon \mathcal{D}_R) (\mathcal{D}_L^T \epsilon \mathcal{D}_L) \\ &= \epsilon_{\alpha\beta} \mathcal{D}_{R\alpha} (\mathcal{D}_L^T \epsilon \mathcal{D}_L) \mathcal{D}_{R\beta}. \end{aligned} \quad (27.2.19)$$

From Eq. (27.2.17) it follows then that W_L and W_R are related by the constraint

$$\epsilon_{\alpha\beta} \mathcal{D}_{L\alpha} W_{L\beta} = \epsilon_{\alpha\beta} \mathcal{D}_{R\alpha} W_{R\beta}. \quad (27.2.20)$$

It is straightforward to show that the most general chiral spinor superfields satisfying Eq. (27.2.20) are of the form (27.2.11) and (27.2.12), with $f_{\mu\nu}$ constrained to satisfy the ‘Bianchi’ identities $\epsilon^{\mu\nu\rho\sigma} \partial_\rho f_{\mu\nu} = 0$.

27.3 Gauge-Invariant Action for General Gauge Superfields

Our experience in the previous section with supersymmetric Abelian gauge theories suggests immediately that in a general non-Abelian gauge theory the kinematic Lagrangian for the fields $V_\mu^A(x)$, $\lambda^A(x)$, and $D^A(x)$ should appear as part of the gauge-invariant generalization of Eq. (27.2.6):

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2} \sum_A (\bar{\lambda}_A \not{D} \lambda_A) + \frac{1}{2} \sum_A D_A D_A. \quad (27.3.1)$$

We are now using a basis for the Lie algebra with totally antisymmetric structure constants, and we are consequently not preserving the distinction between upper and lower group indices, writing all indices A , B , etc. as subscripts. Also, $f_{A\mu\nu}$ is the gauge-covariant field-strength tensor

$$f_{A\mu\nu} = \partial_\mu V_{A\nu} - \partial_\nu V_{A\mu} + \sum_{BC} C_{ABC} V_{B\mu} V_{C\nu}, \quad (27.3.2)$$

and $D_\mu \lambda$ is the gauge-covariant derivative of the gaugino field, which in the adjoint representation is

$$(D_\mu \lambda)_A = \partial_\mu \lambda_A + \sum_{BC} C_{ABC} V_{B\mu} \lambda_C. \quad (27.3.3)$$

The question is: does Eq. (27.3.1) yield a supersymmetric action?

Since the Lagrangian density (27.3.1) is manifestly gauge-invariant, we can test whether the action is supersymmetric in any convenient gauge. To find out whether $\delta \mathcal{L}_{\text{gauge}}$ is a derivative at some point X^μ , it is convenient to adopt a specific version of Wess–Zumino gauge in which $V_A^\mu(X) = 0$. Then at X the changes in the component fields are given by Eqs. (??)–(??) at $x = X$ as

$$\delta V_{A\mu} = (\bar{\alpha} \gamma_\mu \lambda_A), \quad (27.3.4)$$

$$\delta \lambda_A = \left(\frac{1}{4} f_{A\mu\nu} [\gamma^\nu, \gamma^\mu] + i \gamma_5 D_A \right) \alpha, \quad (27.3.5)$$

$$\delta D_A = i (\bar{\alpha} \gamma_5 \not{D} \lambda_A). \quad (27.3.6)$$

(We must set x^μ in these expressions equal to X^μ *after* calculating the change under a supersymmetry transformation, not before.) Also, the non-linear terms in $f_A^{\mu\nu}$ are quadratic in the V s and therefore at $x = X$ have zero variation, so at $x = X$

$$\delta f_{A\mu\nu} = (\bar{\alpha} (\gamma_\nu \partial_\mu - \gamma_\mu \partial_\nu) \lambda_A). \quad (27.3.7)$$

With one exception, the terms in Eq. (27.3.1) and their transformations under supersymmetry transformations are thus just a number of copies (labelled by A) of the Abelian theory discussed in the previous section, and therefore give a supersymmetric action. The one exception that might disturb the supersymmetry of the action arises from the second term in the gauge-covariant derivative (27.3.3) of the gaugino field:

$$\mathcal{L}_{\lambda\lambda V} = -\frac{1}{2} \sum_{ABC} C_{ABC} (\bar{\lambda}_A \not{V}_B \lambda_C), \quad (27.3.8)$$

whose variation at $x = X$ is

$$\begin{aligned}\delta\mathcal{L}_{\lambda\lambda V} &= -\frac{1}{2} \sum_{ABC} C_{ABC} (\bar{\lambda}_A (\delta V_B) \lambda_C) \\ &= -\frac{1}{2} \sum_{ABC} C_{ABC} (\bar{\lambda}_A \gamma_\mu \lambda_C) (\bar{\alpha} \gamma^\mu \lambda_B).\end{aligned}\tag{27.3.9}$$

We can write the product of bilinears on the right-hand side as the sum of two terms

$$(\bar{\lambda}_A \gamma_\mu \lambda_C) (\bar{\alpha} \gamma^\mu \lambda_B) = X_{ABC} + Y_{ABC},$$

with

$$\begin{aligned}X_{ABC} &\equiv \frac{1}{4} \sum_{\pm} (\bar{\lambda}_A (1 \pm \gamma_5) \gamma_\mu \lambda_C) (\bar{\alpha} \gamma^\mu (1 \pm \gamma_5) \lambda_B), \\ Y_{ABC} &\equiv \frac{1}{4} \sum_{\pm} (\bar{\lambda}_A (1 \pm \gamma_5) \gamma_\mu \lambda_C) (\bar{\alpha} \gamma^\mu (1 \mp \gamma_5) \lambda_B).\end{aligned}$$

By using standard Fierz identities and the anticommutativity of the spinor fields, we have

$$\begin{aligned}(\bar{\lambda}_A (1 \pm \gamma_5) \gamma_\mu \lambda_B) (\bar{\alpha} (1 \pm \gamma_5) \gamma^\mu \lambda_C) \\ = (\bar{\lambda}_A (1 \pm \gamma_5) \gamma_\mu \lambda_C) (\bar{\alpha} (1 \pm \gamma_5) \gamma^\mu \lambda_B), \\ (\bar{\lambda}_A (1 \pm \gamma_5) \gamma_\mu \lambda_B) (\bar{\alpha} (1 \mp \gamma_5) \gamma^\mu \lambda_C) \\ = (\bar{\lambda}_C (1 \pm \gamma_5) \gamma_\mu \lambda_B) (\bar{\alpha} (1 \mp \gamma_5) \gamma^\mu \lambda_A).\end{aligned}$$

(To derive the first of these relations, we note that $[(1 \pm \gamma_5) \gamma_\mu]_{\alpha\gamma} [(1 \pm \gamma_5) \gamma^\mu]_{\delta\beta}$ may be thought of as the $\alpha\beta$ matrix element of a matrix depending on δ and γ , and may therefore be expanded in a linear combination of $\mathbf{1}_{\alpha\beta}$, $(\gamma^\mu)_{\alpha\beta}$, $[\gamma^\mu, \gamma^\kappa]_{\alpha\beta}$, $(\gamma_5 \gamma^\mu)_{\alpha\beta}$, and $(\gamma_5)_{\alpha\beta}$. Because of the factors $(1 \pm \gamma_5)$, the only term in the expansion is proportional to $[(1 \pm \gamma_5) \gamma^\mu]_{\alpha\beta}$. Lorentz invariance and the presence of the other $1 \pm \gamma_5$ factor tell us that this expansion takes the form

$$[(1 \pm \gamma_5) \gamma_\mu]_{\alpha\gamma} [(1 \pm \gamma_5) \gamma^\mu]_{\delta\beta} = k [(1 \pm \gamma_5) \gamma_\mu]_{\alpha\beta} [(1 \pm \gamma_5) \gamma^\mu]_{\delta\gamma}.$$

To determine the proportionality constant k , we may contract both sides with $(\gamma_\nu)_{\gamma\alpha}$ and find $k = -1$. The minus sign is cancelled by a minus sign arising from the anticommutation of λ_C and $\bar{\alpha}$. The other Fierz identity is proved in the same way, except that we also need to use the symmetry property (??) for Majorana bilinears.) Hence X_{ABC} is symmetric under interchange of B and C , while Y_{ABC} is symmetric under interchange of A and B . Since C_{ABC} is totally antisymmetric, both X_{ABC} and Y_{ABC} make a vanishing contribution to the sum in Eq. (27.3.9), leaving us $\delta\mathcal{L}_{\lambda\lambda V} = 0$, so that Eq. (27.3.1) gives a supersymmetric action, as was to be shown.

We can understand *why* Eq. (27.3.1) gives a supersymmetric action by identifying the superfield that has $f_{A\mu\nu}$, λ_A , and D_A as component fields. Recall that under a generalized gauge transformation, the vector superfield $V_A(x, \theta)$ has the transformation property

(27.1.12):

$$\begin{aligned} \exp\left(-2\sum_A t_A V_A(x, \theta)\right) &\longrightarrow \exp\left(-i\sum_A t_A \Omega_A(x, \theta)\right) \exp\left(-2\sum_A t_A V_A(x, \theta)\right) \\ &\quad \times \exp\left(+i\sum_A t_A \Omega_A^*(x, \theta)\right), \end{aligned} \quad (27.3.10)$$

where $\Omega_A(x, \theta)$ is a general left-chiral superfield. This is not a gauge-covariant transformation rule, because $\Omega_A^* \neq \Omega_A$. To eliminate the factor involving Ω_A^* , we note that Ω_A^* is a right-chiral superfield, so that $\mathcal{D}_{L\alpha}\Omega_A^* = 0$, and therefore

$$\begin{aligned} &\exp\left(-2\sum_A t_A V_A(x, \theta)\right) \mathcal{D}_{L\alpha} \exp\left(+2\sum_A t_A V_A(x, \theta)\right) \\ &\longrightarrow \exp\left(-i\sum_A t_A \Omega_A(x, \theta)\right) \exp\left(-2\sum_A t_A V_A(x, \theta)\right) \\ &\quad \times \mathcal{D}_{L\alpha} \left[\exp\left(+2\sum_A t_A V_A(x, \theta)\right) \exp\left(+i\sum_A t_A \Omega_A(x, \theta)\right) \right]. \end{aligned} \quad (27.3.11)$$

This is still not gauge-covariant, because the left-superderivative $\mathcal{D}_{L\alpha}$ acts on $\exp(+i\sum_A t_A \Omega_A(x, \theta))$ as well as on $\exp(+2\sum_A t_A V_A(x, \theta))$. This is eliminated if we follow the lead of the Abelian theory discussed in the previous section, and define a spinor superfield

$$\begin{aligned} 2\sum_A t_A W_{AL\alpha}(x, \theta) &\equiv \sum_{\beta\gamma} \epsilon_{\beta\gamma} \mathcal{D}_{R\beta} \mathcal{D}_{R\gamma} \left[\exp\left(-2\sum_A t_A V_A(x, \theta)\right) \right. \\ &\quad \left. \times \mathcal{D}_{L\alpha} \exp\left(+2\sum_A t_A V_A(x, \theta)\right) \right]. \end{aligned} \quad (27.3.12)$$

Because the product of any three $\mathcal{D}_{R\beta}$ s vanishes, $W_{AL\alpha}$ is left-chiral

$$\mathcal{D}_{R\beta} W_{AL\alpha}(x, \theta) = 0, \quad (27.3.13)$$

and because $\mathcal{D}_{R\beta} \mathcal{D}_{R\gamma} \mathcal{D}_{L\alpha} \Omega_A \propto \mathcal{D}_{R\delta} \Omega_A = 0$, $W_{AL\alpha}$ is gauge-covariant in the sense that, for a generalized gauge transformation,

$$\begin{aligned} \sum_A t_A W_{AL\alpha}(x, \theta) &\longrightarrow \exp\left(-i\sum_A t_A \Omega_A(x, \theta)\right) \sum_A t_A W_{AL\alpha}(x, \theta) \\ &\quad \times \exp\left(+i\sum_A t_A \Omega_A(x, \theta)\right). \end{aligned} \quad (27.3.14)$$

To calculate the spinor superfield at a point $x^\mu = X^\mu$, we can again adopt a version of Wess–Zumino gauge in which $V_A(X) = 0$, and after a straightforward calculation find that

in this gauge

$$W_{AL}(X, \theta) = \lambda_{AL}(X_+) + \frac{1}{2} \gamma^\mu \gamma^\nu \theta_L (\partial_\mu V_{A\nu}(X_+) - \partial_\nu V_{A\mu}(X_+)) \\ + (\theta_L^T \epsilon \theta_L) \not{D} \lambda_{RA}(X_+) - i \theta_L D_A(X_+).$$

Since W_{AL} is gauge-covariant, at a general point in a general gauge it must have the value

$$W_{AL}(x, \theta) = \lambda_{AL}(x_+) + \frac{1}{2} \gamma^\mu \gamma^\nu \theta_L f_{A\mu\nu}(x_+) + (\theta_L^T \epsilon \theta_L) \not{D} \lambda_{RA}(x_+) - i \theta_L D_A(x_+). \quad (27.3.15)$$

From this, we can construct a Lorentz- and gauge-invariant $\mathcal{F}$ -term bilinear in W

$$-\left[\sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{AL\alpha} W_{AL\beta} \right]_{\mathcal{F}} = \sum_A \left[-(\bar{\lambda}_A \not{D} (1 - \gamma_5) \lambda_A) - \frac{1}{2} f_{A\mu\nu} f_A^{\mu\nu} \right. \\ \left. + \frac{i}{4} \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_A^{\rho\sigma} + D_A^2 \right]. \quad (27.3.16)$$

Just as in the previous section, the gauge-invariant Lagrangian (27.3.1) is obtained from the real part of this $\mathcal{F}$ -term:

$$-\frac{1}{2} \text{Re} \left[\sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{AL\alpha} W_{AL\beta} \right]_{\mathcal{F}} = \mathcal{L}_{\text{gauge}}. \quad (27.3.17)$$

What about the imaginary part? This is given by

$$-\text{Im} \left[\sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{AL\alpha} W_{AL\beta} \right]_{\mathcal{F}} = -i \sum_A (\bar{\lambda}_A \not{D} \gamma_5 \lambda_A) \\ + \frac{1}{4} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma}. \quad (27.3.18)$$

Eq. (??) and the antisymmetry of the structure constants show that $(\bar{\lambda}_A \not{D} \gamma_5 \lambda_A) = \frac{1}{2} \partial_\mu (\bar{\lambda}_A \gamma^\mu \gamma_5 \lambda_A)$, so the first term is a total derivative, while Eq. (??) tells us that the second term is a total derivative also. In Abelian gauge theories, this means that a term like (27.3.18) would have no effect. But as discussed in Sections 23.5 and 23.6, in non-Abelian gauge theories the existence of instanton solutions allows the density (27.3.18) to have a non-vanishing integral over spacetime. We therefore must allow for the possibility of a new term in the Lagrangian density

$$\mathcal{L}_\theta = -\frac{g^2 \theta}{16\pi^2} \text{Im} \left[\sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{AL\alpha} W_{AL\beta} \right]_{\mathcal{F}}, \quad (27.3.19)$$

where θ is a new real parameter, and g is a gauge coupling, which can be conveniently defined for a simple gauge group so that if t_A , t_B , and t_C are in the ‘standard’ $SU(2)$ subalgebra of the gauge algebra used in calculating instanton effects, we have $C_{ABC} = g\epsilon_{ABC}$. With this definition of the gauge coupling, Eq. (??) gives for simple gauge groups

$$\int d^4x \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma} = 64\pi^2 \nu / g^2, \quad (27.3.20)$$

where $\nu = 0, \pm 1, \pm 2, \dots$ is an integer, the winding number, which characterizes the topological class of the gauge field configuration. Thus for instantons of winding number ν , the Lagrangian density $\mathcal{L}_\theta$ contributes a phase to path integrals, given by

$$\left[\exp \left(i \int d^4x \mathcal{L}_\theta \right) \right]_\nu = \exp(i\nu\theta), \quad (27.3.21)$$

so the effects of $\mathcal{L}_\theta$ are periodic in θ , with period 2π .

It is often convenient to absorb a factor g into the gauge field, so that the structure constants do not depend on g , and instead the Lagrangian density for the gauge field is multiplied by an over-all factor $1/g^2$. In this notation, the complete Lagrangian density for the gauge field may be written in terms of the rescaled gauge fields and structure constants as

$$\mathcal{L}_{\text{gauge}} + \mathcal{L}_\theta = -\text{Re} \left[\frac{\tau}{8\pi i} \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{AL\alpha} W_{AL\beta} \right]_{\mathcal{F}}, \quad (27.3.22)$$

where τ is the complex coupling parameter

$$\tau = \frac{4\pi i}{g^2} + \frac{\theta}{2\pi}. \quad (27.3.23)$$

According to Eq. (??), the contribution of instantons of winding number ν to path integrals is suppressed by a factor $\exp(-8\pi^2|\nu|/g^2)$, which together with the factor (27.3.21) yields an over-all factor

$$\exp \left[i\nu\theta - \frac{8\pi^2|\nu|}{g^2} \right] = \begin{cases} \exp(2\pi i\nu\tau), & \nu \geq 0, \\ \exp(2\pi i\nu\tau^*), & \nu \leq 0. \end{cases} \quad (27.3.24)$$

27.4 Renormalizable Gauge Theories with Chiral Superfields

We will now put together the pieces assembled in the previous three sections, to construct the most general renormalizable action for chiral superfields interacting with general gauge fields. Adding the terms (27.1.27), (27.2.7), and (27.3.1) and the superpotential terms in

Eq. (??) gives the Lagrangian density

$$\begin{aligned}
\mathcal{L} &= \frac{1}{2} \left[\Phi^\dagger \exp \left(-2 \sum_A t_A V_A \right) \Phi \right]_D - \frac{1}{2} \text{Re} \sum_A (W_{AL}^T \epsilon W_{AL})_{\mathcal{F}} \\
&\quad - \frac{g^2 \theta}{16\pi^2} \sum_A \text{Im} (W_{AL}^T \epsilon W_{AL})_{\mathcal{F}} - \sum_A \xi_A [V_A]_D + 2 \text{Re}[f]_{\mathcal{F}} \\
&= - \sum_n (D_\mu \phi)_n^* (D^\mu \phi)_n - \frac{1}{2} \sum_n (\bar{\psi}_n \gamma^\mu (D_\mu \psi_L)_n) + \sum_n \mathcal{F}_n^* \mathcal{F}_n \\
&\quad - \text{Re} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} (\psi_{nL}^T \epsilon \psi_{mL}) + 2 \text{Re} \sum_n \frac{\partial f(\phi)}{\partial \phi_n} \mathcal{F}_n \\
&\quad - 2\sqrt{2} \text{Im} \sum_{Anm} (t_A)_{nm} (\bar{\psi}_{nL} \lambda_A) \phi_m + 2\sqrt{2} \text{Im} \sum_{Anm} (t_A)_{mn} (\bar{\psi}_{nR} \lambda_A) \phi_m^* \\
&\quad - \sum_{Anm} \phi_n^* (t_A)_{nm} \phi_m D_A - \sum_A \xi_A D_A + \frac{1}{2} \sum_A D_A D_A \\
&\quad - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma}.
\end{aligned} \tag{27.4.1}$$

Here $f(\phi)$ is the superpotential, a gauge-invariant complex function of ϕ_n (but not of ϕ_n^*) which the condition of renormalizability requires to be a cubic polynomial; the ξ_A are constants which gauge invariance requires to vanish except where t_A is a $U(1)$ generator; the gauge-covariant derivatives are

$$D_\mu \psi_L \equiv \partial_\mu \psi_L - i \sum_A t_A V_{A\mu} \psi_L, \tag{27.4.2}$$

$$D_\mu \phi \equiv \partial_\mu \phi - i \sum_A t_A V_{A\mu} \phi, \tag{27.4.3}$$

$$(D_\mu \lambda)_A = \partial_\mu \lambda_A + \sum_{BC} C_{ABC} V_{B\mu} \lambda_C, \tag{27.4.4}$$

and $f_{A\mu\nu}$ is the gauge-covariant gauge field-strength tensor

$$f_{A\mu\nu} = \partial_\mu V_{A\nu} - \partial_\nu V_{A\mu} + \sum_{BC} C_{ABC} V_{B\mu} V_{C\nu}. \tag{27.4.5}$$

The auxiliary fields enter quadratically, with field-independent constants as the coefficients of the second-order term, so they can be eliminated by setting them equal to the values at which the Lagrangian density is stationary:

$$\mathcal{F}_n = - \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^*, \tag{27.4.6}$$

$$D_A = \xi_A + \sum_{nm} \phi_n^* (t_A)_{nm} \phi_m. \tag{27.4.7}$$

Using these back in Eq. (27.4.1), the Lagrangian density becomes

$$\begin{aligned}
\mathcal{L} = & - \sum_n (D_\mu \phi)_n^* (D^\mu \phi)_n - \frac{1}{2} \sum_n (\bar{\psi}_{nL} \gamma^\mu (D_\mu \psi_L)_n) + \frac{1}{2} \sum_n \left(\overline{(D_\mu \psi_L)_n} \gamma^\mu \psi_{nL} \right) \\
& - \frac{1}{2} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} (\psi_{nL}^\text{T} \epsilon \psi_{mL}) - \frac{1}{2} \sum_{nm} \left(\frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} \right)^* (\psi_{nL}^\text{T} \epsilon \psi_{mL})^* \\
& - \sum_n \left| \frac{\partial f(\phi)}{\partial \phi_n} \right|^2 + i\sqrt{2} \sum_{Anm} (\bar{\psi}_{nL}(t_A)_{nm} \lambda_A) \phi_m \\
& - i\sqrt{2} \sum_{Anm} \phi_n^* (\bar{\lambda}_A(t_A)_{nm} \psi_{mL}) - \frac{1}{2} \sum_A \left(\xi_A + \sum_{nm} \phi_n^*(t_A)_{nm} \phi_m \right)^2 \\
& - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma}.
\end{aligned} \tag{27.4.8}$$

Lorentz invariance requires the fields ψ_{nL} , λ_A , and $f_{A\mu\nu}$ to have vanishing vacuum expectation values, while the tree-value vacuum expectation values of the ϕ_n are at the minimum of the potential

$$V(\phi) = \sum_n \left| \frac{\partial f(\phi)}{\partial \phi_n} \right|^2 + \frac{1}{2} \sum_A \left(\xi_A + \sum_{nm} \phi_n^*(t_A)_{nm} \phi_m \right)^2. \tag{27.4.9}$$

This potential is positive, so if there is a set of field values at which $V(\phi)$ vanishes, then this is automatically also a minimum of the potential. For $V(\phi)$ to vanish at some field value $\phi_n = \phi_{n0}$, it is necessary and sufficient that

$$\mathcal{F}_{n0} = - \left[\frac{\partial f(\phi)}{\partial \phi_n} \right]_{\phi=\phi_0}^* = 0 \tag{27.4.10}$$

and

$$D_{A0} = \xi_A + \sum_{nm} \phi_{n0}^*(t_A)_{nm} \phi_{m0} = 0. \tag{27.4.11}$$

This in turn is the necessary and sufficient condition for supersymmetry not to be spontaneously broken, since Eq. (??) gives $\langle \delta \psi_{nL} \rangle_{\text{VAC}} = \sqrt{2} \langle \mathcal{F}_n \rangle_{\text{VAC}} \alpha_L$, and Eq. (??) gives $\langle \delta \lambda_A \rangle_{\text{VAC}} = i \langle D_A \rangle_{\text{VAC}} \gamma_5 \alpha$.

It is worth stressing here that spontaneous symmetry breaking is more difficult for supersymmetry than for other symmetries. For most symmetries of the action there will be field configurations at which the symmetry is unbroken and the potential is stationary, but the symmetry will nevertheless be spontaneously broken if none of these configurations are minima of the potential. In contrast, any supersymmetric field configuration gives the potential a value of zero, which is necessarily lower than the value of the potential for any non-supersymmetric configuration, so the existence of *any* supersymmetric field configuration insures that supersymmetry is unbroken. As we will see in Section 27.6, this conclusion goes beyond the tree approximation used in this section; it is unaffected by corrections of any finite order in perturbation theory.

It may appear that Eqs. (27.4.10) and (27.4.11) impose too many conditions on the scalar fields to expect a solution, without some fine-tuning of the superpotential. However, for a gauge group of dimensionality D , the superpotential $f(\phi)$ is subject to the D constraints

$$\sum_m \frac{\partial f(\phi)}{\partial \phi_m} (t_A \phi)_m = 0, \quad (27.4.12)$$

for all A and all ϕ . Hence if ϕ has N independent components, then the number of *independent* conditions (27.4.10) is $N - D$, while the number of conditions (27.4.11) is D , so there are just N conditions altogether. With the number of conditions equal to the number of free variables, it is likely to find solutions for generic superpotentials. In fact, it is more usual to find solutions than not. For instance, for chiral scalar superfields in a non-trivial representation of a semi-simple gauge group we have $\xi_A = 0$, while $f(\phi)$ can have no terms linear in the ϕ_n , so both Eqs. (27.4.10) and (27.4.11) are satisfied for $\phi_{n0} = 0$. There may be other solutions of Eqs. (27.4.10) and (27.4.11) which break the gauge symmetries, but in such a theory supersymmetry cannot be broken, at least not in the tree approximation, and, as we shall see in Section 27.6, not in any order of perturbation theory.

More generally, it is easy to see that, even if the gauge group has $U(1)$ factors and even if the superpotential involves gauge-invariant superfields, if there exists a set of scalar field values ϕ_{n0} that satisfy Eq. (27.4.10), then there is another that satisfies both Eq. (27.4.10) and Eq. (27.4.11), provided only that the Fayet–Iliopoulos constants ξ_A all vanish. To show this, we note that since the superpotential $f(\phi)$ does not involve ϕ^* , it is invariant not only under ordinary gauge transformations $\phi \rightarrow \exp(i \sum_A \Lambda_A t_A) \phi$, with Λ_A arbitrary real numbers, but also under transformations with Λ_A arbitrary complex numbers. Under all these transformations the $\mathcal{F}$ -terms in Eq. (27.4.10) transform linearly, so if ϕ_0 satisfies Eq. (27.4.10), then so does $\phi^\Lambda \equiv \exp(i \sum_A \Lambda_A t_A) \phi_0$. On the other hand, the scalar product $[\phi^\dagger \phi]$ is not invariant under transformations with Λ_A complex, but $[\phi^{\Lambda\dagger} \phi^\Lambda]$ does remain real and positive for complex Λ_A , so it is bounded below, and therefore has a minimum. For $\xi_A = 0$, the condition that $[\phi^{\Lambda\dagger} \phi^\Lambda]$ be stationary at this minimum is just that ϕ^Λ should satisfy Eq. (27.4.11). We see then that in the absence of Fayet–Iliopoulos D -terms, the question of whether supersymmetry is unbroken in gauge theories is entirely a matter of whether the superpotential allows solutions of Eq. (27.4.10). The same result applies even in non-renormalizable theories [3].

Now let us assume that there is a set of values ϕ_{n0} at which $V(\phi_0) = 0$, so that supersymmetry is unbroken. The spin 0 degrees of freedom are described by a shifted field

$$\varphi_n = \phi_n - \phi_{n0}. \quad (27.4.13)$$

There is then a cross-term between φ and the gauge fields, arising from the first term in Eq. (27.4.1):

$$2 \sum_{nA} \text{Im} (\partial_\mu \varphi_n (t_A \phi_0)_n^*) V_A^\mu.$$

As shown in Section 21.1, it is always possible to eliminate this term by adopting a ‘unitarity

gauge,' in which ϕ_n satisfies a constraint that makes this term vanish:

$$\sum_n \text{Im}(\phi_n(t_A\phi_0)_n^*) = 0. \quad (27.4.14)$$

This will have the effect of eliminating the Goldstone bosons associated with broken gauge symmetries.

We shall now work out the masses of the particles of spin 0, spin 1/2, and spin 1 that arise in this theory if supersymmetry is unbroken, taking into account the possibility that the gauge symmetries may be spontaneously broken.

Spin 0

Because $\partial f(\phi)/\partial\phi_n$ and $\xi_A + \sum_{nm} \phi_n^*(t_A)_{nm}\phi_m$ must both vanish at $\phi_n = \phi_{n0}$, the terms in $V(\phi)$ of second order in $\varphi_n \equiv \phi_n - \phi_{n0}$ and/or φ_n^* are of the form

$$\begin{aligned} V_{\text{quad}}(\phi) = & \sum_{nm} (\mathcal{M}^* \mathcal{M})_{nm} \varphi_n^* \varphi_m + \sum_{Anm} (t_A\phi_0)_n (t_A\phi_0)_m^* \varphi_n^* \varphi_m \\ & + \frac{1}{2} \sum_{Anm} (t_A\phi_0)_n^* (t_A\phi_0)_m^* \varphi_n \varphi_m + \frac{1}{2} \sum_{Anm} (t_A\phi_0)_n (t_A\phi_0)_m \varphi_n^* \varphi_m^*, \end{aligned} \quad (27.4.15)$$

where $\mathcal{M}$ is the complex symmetric matrix (??):

$$\mathcal{M}_{nm} \equiv \left(\frac{\partial^2 f(\phi)}{\partial\phi_n \partial\phi_m} \right)_{\phi=\phi_0}.$$

This can be written as

$$V_{\text{quad}} = \frac{1}{2} \begin{bmatrix} \varphi \\ \varphi^* \end{bmatrix}^\dagger M_0^2 \begin{bmatrix} \varphi \\ \varphi^* \end{bmatrix}, \quad (27.4.16)$$

where M_0^2 is the block matrix

$$M_0^2 = \begin{bmatrix} \mathcal{M}^* \mathcal{M} + \sum_A (t_A\phi_0)(t_A\phi_0)^\dagger & \sum_A (t_A\phi_0)(t_A\phi_0)^T \\ \sum_A (t_A\phi_0)^*(t_A\phi_0)^\dagger & \mathcal{M} \mathcal{M}^* + \sum_A (t_A\phi_0)^*(t_A\phi_0)^T \end{bmatrix}. \quad (27.4.17)$$

Now we must look for the eigenvalues of this mass-squared matrix. Differentiating Eq. (27.4.12) with respect to ϕ_n gives

$$\sum_m \frac{\partial^2 f(\phi)}{\partial\phi_n \partial\phi_m} (t_A\phi)_m + \sum_m \frac{\partial f(\phi)}{\partial\phi_m} (t_A)_{mn} = 0. \quad (27.4.18)$$

But as we have seen, $\partial f(\phi)/\partial\phi_m$ vanishes at $\phi = \phi_0$, so by setting ϕ at this value in Eq. (27.4.18), we find

$$\sum_m \mathcal{M}_{nm} (t_A\phi_0)_m = 0. \quad (27.4.19)$$

It follows that

$$M_0^2 \begin{bmatrix} t_B\phi_0 \\ \pm(t_B\phi_0)^* \end{bmatrix} = \sum_A \left(\phi_0^\dagger [t_A t_B \pm t_B t_A] \phi_0 \right) \begin{bmatrix} t_B\phi_0 \\ \pm(t_B\phi_0)^* \end{bmatrix}.$$

But the vanishing of D_A at $\phi = \phi_0$ and the global gauge invariance of ξ_A tell us that

$$\left(\phi_0^\dagger[t_A, t_B]\phi_0\right) = i \sum_C C_{ABC} \left(\phi_0^\dagger t_C \phi_0\right) = -i \left(\phi_0^\dagger \phi_0\right) \sum_C C_{ABC} \xi_C = 0. \quad (27.4.20)$$

The matrix (27.4.17) therefore has a pair of eigenvectors for each gauge symmetry

$$u = \begin{bmatrix} \sum_B c_B t_B \phi_0 \\ \sum_B c_B (t_B \phi_0)^* \end{bmatrix}, \quad v = \begin{bmatrix} \sum_B c_B t_B \phi_0 \\ -\sum_B c_B (t_B \phi_0)^* \end{bmatrix}, \quad (27.4.21)$$

for which

$$M_0^2 u = \mu^2 u, \quad M_0^2 v = 0, \quad (27.4.22)$$

where μ^2 and c_A are any real solutions of the eigenvalue problem²

$$\sum_B \left(\phi_0^\dagger \{t_A, t_B\} \phi_0\right) c_B = \mu^2 c_A, \quad (27.4.23)$$

with the exception that if the eigenvalue μ^2 vanishes then $\sum_B c_B t_B \phi_0 = 0$, so that the eigenvectors u and v are absent. The massless particles associated with the v eigenvectors are Goldstone bosons, which are eliminated from the physical spectrum by the unitarity gauge condition (27.4.14). In addition to these mass eigenstates, there is another set that are orthogonal to all the u s and v s and that therefore take the form

$$w_\pm = \begin{bmatrix} \zeta \\ \pm \zeta^* \end{bmatrix}, \quad (27.4.24)$$

where

$$\sum_n (t_A \phi_0)_n^* \zeta_n = 0. \quad (27.4.25)$$

Eq. (27.4.19) shows that the space of ζ s satisfying Eq. (27.4.25) is invariant under multiplication with the Hermitian matrix $\mathcal{M}^\dagger \mathcal{M}$, so it is spanned by the eigenvectors of this matrix, satisfying

$$\mathcal{M}^\dagger \mathcal{M} \zeta = m^2 \zeta \quad (27.4.26)$$

with m^2 a set of real positive (or zero) eigenvalues. Eq. (27.4.26) and its complex conjugate together with Eq. (27.4.25) show that $w_\pm$ are eigenvectors of M_0^2 with eigenvalues m^2 :

$$M_0^2 w_\pm = m^2 w_\pm. \quad (27.4.27)$$

We thus have *two* self-charge-conjugate spinless bosons of each mass m satisfying Eq. (27.4.27), and one self-charge-conjugate spinless boson of each non-zero mass μ satisfying Eq. (27.4.23).

²The presence of a factor 1/2 in Eq. (??) which does not appear in Eq. (27.4.23) is due to a difference in the way that the scalar fields are normalized.

Spin 1/2

The fermion masses arise from the non-derivative terms in Eq. (27.4.8) that are of second order in the fermion fields ψ_n and λ_A :

$$\mathcal{L}_{1/2} = -\frac{1}{2} \sum_{nm} \mathcal{M}_{nm} (\psi_{nL}^T \epsilon \psi_{mL}) - i\sqrt{2} \sum_{Am} (t_A \phi_0)_m^* (\lambda_{LA}^T \epsilon \psi_{mL}) + \text{H.c.} \quad (27.4.28)$$

We saw in Section 26.4 that if the fermion mass term in the Lagrangian for a column χ of Majorana spinor fields is put in the form

$$\mathcal{L}_{1/2} = -\frac{1}{2} (\chi_L^T \epsilon M \chi_L) + \text{H.c.}, \quad (27.4.29)$$

then the fermion squared masses are the eigenvalues of the Hermitian matrix $M^\dagger M$. Eq. (27.4.28) gives the elements of the matrix M here as

$$M_{nm} = \mathcal{M}_{nm}, \quad M_{nA} = M_{An} = i\sqrt{2} (t_A \phi_0)_n^*, \quad M_{AB} = 0, \quad (27.4.30)$$

for which, using Eqs. (27.4.19) and (27.4.20),

$$\begin{aligned} (M^\dagger M)_{nm} &= (\mathcal{M}^\dagger \mathcal{M})_{nm} + 2 \sum_A (t_A \phi_0)_n (t_A \phi_0)_m^*, \\ (M^\dagger M)_{nA} &= (M^\dagger M)_{An} = 0, \\ (M^\dagger M)_{AB} &= 2 (\phi_0^\dagger t_B t_A \phi_0) = (\phi_0^\dagger \{t_B, t_A\} \phi_0). \end{aligned} \quad (27.4.31)$$

The eigenvectors of the matrix (27.4.30) are of three types. First are those of the form

$$z = \begin{bmatrix} \zeta \\ 0 \end{bmatrix}, \quad (27.4.32)$$

with eigenvalues m^2 , where ζ_n and m^2 are any eigenvectors and corresponding eigenvalues of $\mathcal{M}^\dagger \mathcal{M}$. Next are those of the form

$$g = \begin{bmatrix} 0 \\ c \end{bmatrix}, \quad (27.4.33)$$

with eigenvalues μ^2 , where c_B and μ^2 are any eigenvectors and eigenvalues of the matrix $(\phi_0^\dagger \{t_B, t_A\} \phi_0)$. Finally, there are those of the form

$$h = \begin{bmatrix} \sum_B c_B t_B \phi_0 \\ 0 \end{bmatrix}, \quad (27.4.34)$$

with eigenvalues μ^2 , where c_B and μ^2 are again any eigenvectors and eigenvalues of the matrix $(\phi_0^\dagger \{t_B, t_A\} \phi_0)$. The only exception is that the eigenvectors c of this matrix with eigenvalue zero have $\sum_A c_A t_A \phi_0 = 0$, corresponding to unbroken symmetries, so that in this case the vector (27.4.34) vanishes and we have only the eigenvector (27.4.33). Thus there is one Majorana fermion of each mass m satisfying Eq. (27.4.26), two Majorana fermions of each non-zero mass μ satisfying Eq. (27.4.22), and one Majorana fermion of zero mass for each unbroken gauge symmetry.

Spin 1

The mass terms in the Lagrangian for the gauge fields arise from the part of the first term in Eq. (27.4.1) that is of second order in the gauge field V_A^μ :

$$\mathcal{L}_V = - \sum_{nAB} (t_A \phi_0)_n^* (t_B \phi_0)_n V_{A\mu} V_B^\mu. \quad (27.4.35)$$

Since the fields $V_{A\mu}$ are real, their mass-squared matrix is the matrix in Eq. (27.4.23):

$$(\mu^2)_{AB} = \left(\phi_0^\dagger \{t_B, t_A\} \phi_0 \right). \quad (27.4.36)$$

There is one spin 1 particle of mass μ for each eigenvalue μ^2 of the matrix (27.4.36).

Putting this all together, we see that for each eigenvalue m^2 of the matrix $\mathcal{M}^* \mathcal{M}$ there are two self-charge-conjugate spinless particles and one Majorana fermion of mass m ; for each non-zero eigenvalue of the matrix μ_{AB}^2 there are one self-charge-conjugate spinless boson, two Majorana fermions, and one self-charge-conjugate spin 1 boson of mass μ ; and for each zero eigenvalue of this matrix there is one Majorana fermion and one self-charge-conjugate spin 1 boson of zero mass. It is not surprising that the particle multiplets for each zero or non-zero mass are just the same as we found by direct use of the supersymmetry algebra in Sections 25.4 and 25.5. What is a bit surprising is that the masses of the gauge and chiral particles are unaffected by each other. The masses m that are given by eigenvalues of $(\mathcal{M}^* \mathcal{M})_{nm}$ and the particles with these masses are just what they would be in a theory of chiral superfields without gauge superfields, and the masses μ that are given by eigenvalues of μ_{AB}^2 and the particles with these masses are just what they would be in a theory of gauge superfields with no chiral superfields.

For future use in Section 27.9 we will now apply the method described in Section 26.7 to construct the supersymmetry current for the supersymmetric gauge Lagrangian (27.4.1). In the gauge used earlier, an infinitesimal supersymmetry transformation changes V_A , λ_A , and D_A by the amounts (27.3.4)–(27.3.6). Adding the Noether supersymmetry current given by Eq. (??) for these fields to the Noether current for ϕ_n , ψ_n , and $\mathcal{F}_n$ already given in Eq. (??), with derivatives replaced by gauge-invariant derivatives, gives the total Noether supersymmetry current:

$$\begin{aligned} N^\mu = & \sum_A f_A^{\mu\nu} \gamma_\nu \lambda_A - \frac{1}{8} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu \lambda_A - \frac{1}{2} i \sum_A D_A \gamma_5 \gamma^\mu \lambda_A \\ & + \frac{1}{\sqrt{2}} \sum_n [2(D^\mu \phi)_n^* \psi_{nL} + 2(D^\mu \phi)_n \psi_{nR} + (\not{D}\phi)_n \gamma^\mu \psi_{nR} \\ & + (\not{D}\phi)_n^* \gamma^\mu \psi_{nL} - \mathcal{F}_n \gamma^\mu \psi_{nR} - \mathcal{F}_n^* \gamma^\mu \psi_{nL}]. \end{aligned} \quad (27.4.37)$$

This is not the supersymmetry current, because the Lagrangian density is not invariant under supersymmetry; instead, its change is the derivative

$$\delta \mathcal{L} = \partial_\mu (\bar{\alpha} K^\mu), \quad (27.4.38)$$

where³

$$\begin{aligned}
K^\mu = & \frac{1}{2}i \sum_A \epsilon^{\rho\sigma\mu\nu} f_{A\rho\sigma} \gamma_\nu \gamma_5 \lambda_A + \frac{1}{8} \sum_A [\gamma^\rho, \gamma^\sigma] \gamma^\mu \lambda_A f_{A\rho\sigma} + \frac{1}{2}i \sum_A D_A \gamma_5 \gamma^\mu \lambda_A \\
& - i \sum_{Anm} (t_A)_{nm} \gamma_5 \gamma^\mu \lambda_A \phi_n^* \phi_m \\
& + \frac{1}{\sqrt{2}} \sum_n \gamma^\mu \left[-(\not{D}\phi)_n \psi_{nR} - (\not{D}\phi)_n^* \psi_{nL} + \mathcal{F}_n^* \psi_{nL} + \mathcal{F}_n \psi_{nR} \right. \\
& \quad \left. + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^* \psi_{nL} + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right) \psi_{nR} \right].
\end{aligned} \tag{27.4.39}$$

The first two terms are derived using the identity (27.2.5). Using the same identity again with Eq. (??) gives the total supersymmetry current:

$$\begin{aligned}
S^\mu = N^\mu + K^\mu = & -\frac{1}{4} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu \lambda_A - i \sum_{Anm} (t_A)_{nm} \gamma_5 \gamma^\mu \lambda_A \phi_n^* \phi_m \\
& + \frac{1}{\sqrt{2}} \sum_n \left[(\not{D}\phi)_n \gamma^\mu \psi_{nR} + (\not{D}\phi)_n^* \gamma^\mu \psi_{nL} \right. \\
& \quad \left. + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right) \gamma^\mu \psi_{nL} + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^* \gamma^\mu \psi_{nR} \right].
\end{aligned} \tag{27.4.40}$$

* * *

In Section 26.8 we considered a class of supersymmetric theories with a superpotential $f(\Phi)$ that has an arbitrary dependence on a set of left-chiral scalar superfields Φ_n but not their derivatives, and with a Kahler potential $K(\Phi, \Phi^*)$ that has an arbitrary dependence on the Φ_n and Φ_n^* , but not their derivatives. We can extend the same considerations to gauge theories, again with the dependence of the Lagrangian on the chiral superfields limited only by supersymmetry, but without introducing new superderivatives or spacetime derivatives. The renormalizable Lagrangian density is then replaced with

$$\begin{aligned}
\mathcal{L} = & \frac{1}{2} \left[K \left(\Phi, \Phi^\dagger \exp \left(-2 \sum_A t_A V_A \right) \right) \right]_D + 2 \operatorname{Re}[f(\Phi)]_{\mathcal{F}} \\
& - \frac{1}{2} \operatorname{Re} \sum_{AB} [h_{AB}(\Phi) (W_{AL}^T \epsilon W_{BL})]_{\mathcal{F}},
\end{aligned} \tag{27.4.41}$$

where $h_{AB}(\Phi)$ is a new function of the Φ_n , but not of the Φ_n^* or derivatives.

The chiral gauge and scalar superfields are given by the expansions (??) and (27.3.15):

$$\begin{aligned}
W_{AL}(x, \theta) = & \lambda_{AL}(x_+) + \frac{1}{2} \gamma^\mu \gamma^\nu \theta_L f_{A\mu\nu}(x_+) + (\theta_L^T \epsilon \theta_L) \not{\partial} \lambda_{AR}(x_+) - i \theta_L D_A(x_+), \\
\Phi_n(x, \theta) = & \phi_n(x_+) - \sqrt{2} (\theta_L^T \epsilon \psi_{nL}(x_+)) + \mathcal{F}_n(x_+) (\theta_L^T \epsilon \theta_L),
\end{aligned}$$

³The easiest way to calculate the change in the term $[\Phi^\dagger \exp(-2 \sum_A t_A V_A) \Phi]_D$ is to calculate the λ -component of $\Phi^\dagger \exp(-2 \sum_A t_A V_A) \Phi$ and use Eq. (??). In this way of doing the calculation, the important term on the second line of the right-hand side of Eq. (27.4.39) arises from the λ -component of $\exp(-2 \sum_A t_A V_A)$.

where x_+^μ is the shifted coordinate (??). The terms of second order in θ_L (and independent of θ_R) in $\sum_{AB} h_{AB}(\Phi)(W_{AL}^T \epsilon W_{BL})$ are then

$$\begin{aligned}
-\left[\sum_{AB} h_{AB}(\Phi) (W_{AL}^T \epsilon W_{BL}) \right]_{\theta_L^2} &= (\theta_L^T \epsilon \theta_L) \sum_{AB} (\lambda_{AL}^T \epsilon \lambda_{BL}) \left[\frac{1}{2} \sum_{nm} (\psi_{nL}^T \epsilon \psi_{mL}) \frac{\partial^2 h_{AB}(\phi)}{\partial \phi_n \partial \phi_m} \right. \\
&\quad \left. - \sum_n \mathcal{F}_n \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \right] \\
&\quad + (\theta_L^T \epsilon \theta_L) \sum_{AB} h_{AB}(\phi) \left[-(\bar{\lambda}_A \not{\partial} (1 - \gamma_5) \lambda_B) - \frac{1}{2} f_{A\mu\nu} f_B^{\mu\nu} \right. \\
&\quad \left. + \frac{i}{4} \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_B^{\rho\sigma} + D_A D_B \right] \\
&\quad + \sqrt{2} \sum_{ABn} \frac{\partial h_{AB}(\phi)}{\partial \phi_n} (\theta_L^T \epsilon \psi_{nL}) \left[-(\lambda_{BL}^T \epsilon \gamma^\mu \gamma^\nu \theta_L) f_{A\mu\nu} \right. \\
&\quad \left. + 2i (\lambda_{BL}^T \epsilon \theta_L) \right],
\end{aligned}$$

with all fields now understood to be evaluated at x^μ rather than x_+^μ . (The first and second terms on the right-hand side are taken from Eqs. (??) and (27.3.16), respectively.) Also, by writing $\theta_{L\alpha} \theta_{L\beta}$ as $\frac{1}{2} \epsilon_{\alpha\beta} (\theta_L^T \epsilon \theta_L)$, the third term on the right-hand side can also be expressed as proportional to $(\theta_L^T \epsilon \theta_L)$:

$$\begin{aligned}
&(\theta_L^T \epsilon \psi_{nL}) [(\bar{\psi}_B \gamma^\mu \gamma^\nu \theta_L) f_{A\mu\nu} - 2i (\bar{\psi}_B \theta_L)] \\
&= \frac{1}{2} (\theta_L^T \epsilon \theta_L) [(\bar{\psi}_B \gamma^\mu \gamma^\nu \psi_{nL}) - 2i (\bar{\psi}_B \psi_{nL}) D_A].
\end{aligned}$$

The $\mathcal{F}$ -term is the coefficient of $(\theta_L^T \epsilon \theta_L)$, so

$$\begin{aligned}
-\left[\sum_{AB} h_{AB}(\Phi) (W_{AL}^T \epsilon W_{BL}) \right]_{\mathcal{F}} &= \sum_{AB} (\lambda_{AL}^T \epsilon \lambda_{BL}) \left[\frac{1}{2} \sum_{nm} (\psi_{nL}^T \epsilon \psi_{mL}) \frac{\partial^2 h_{AB}(\phi)}{\partial \phi_n \partial \phi_m} \right. \\
&\quad \left. - \sum_n \mathcal{F}_n \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \right] \\
&\quad + \sum_{AB} h_{AB}(\phi) \left[-(\bar{\lambda}_A \not{\partial} (1 - \gamma_5) \lambda_B) - \frac{1}{2} f_{A\mu\nu} f_B^{\mu\nu} \right. \\
&\quad \left. + \frac{i}{4} \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_B^{\rho\sigma} + D_A D_B \right] \\
&\quad + \frac{\sqrt{2}}{2} \sum_{ABn} \frac{\partial h_{AB}(\phi)}{\partial \phi_n} [-(\bar{\psi}_B \gamma^\mu \gamma^\nu \psi_{nL}) f_{A\mu\nu} + 2i (\bar{\psi}_B \psi_{nL}) D_A].
\end{aligned}$$

The other terms in Eq. (27.4.41) are just given by the gauge-invariant version of the La-

grangian density (??). Putting this together gives the Lagrangian density

$$\begin{aligned}
\mathcal{L} = & \operatorname{Re} \sum_{nm} \mathcal{G}_{nm}(\phi, \phi^*) \left[-\frac{1}{2} (\bar{\psi}_m \not{D} (1 + \gamma_5) \psi_n) + \mathcal{F}_n \mathcal{F}_m^* - D_\mu \phi_n D^\mu \phi_m^* \right] \\
& - 2 \operatorname{Re} \sum_i \frac{\partial K(\phi, \phi^*)}{\partial \phi_i^*} D_A(\phi^* t_A)_i \\
& + i\sqrt{2} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_i \partial \phi_j^*} [(t_A \phi)_i \bar{\psi}_j \lambda_{AR} - (\phi^* t_A)_j \bar{\psi}_i \lambda_{AL}] \\
& - \operatorname{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_n \psi_{mL}) \mathcal{F}_\ell^* \\
& + \operatorname{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_m \gamma^\mu \psi_{\ell R}) D_\mu \phi_n \\
& + \frac{1}{4} \sum_{nm\ell k} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^* \partial \phi_k^*} (\bar{\psi}_n \psi_{mL}) (\bar{\psi}_k \psi_{\ell R}) \\
& - \operatorname{Re} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} (\bar{\psi}_n \psi_{mL}) + 2 \operatorname{Re} \sum_n \mathcal{F}_n \frac{\partial f(\phi)}{\partial \phi_n} \\
& + \frac{1}{4} \operatorname{Re} \sum_{ABnm} (\bar{\lambda}_A \lambda_{BL}) (\bar{\psi}_n \psi_{mL}) \frac{\partial^2 h_{AB}(\phi)}{\partial \phi_n \partial \phi_m} \\
& - \frac{1}{2} \operatorname{Re} \sum_{ABn} (\bar{\lambda}_A \lambda_{BL}) \mathcal{F}_n \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \\
& + \operatorname{Re} \sum_{AB} h_{AB}(\phi) \left[-(\bar{\lambda}_A \not{D} \lambda_{BR}) - \frac{1}{4} f_{A\mu\nu} f_B^{\mu\nu} + \frac{i}{8} \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_B^{\rho\sigma} + \frac{1}{2} D_A D_B \right] \\
& + \frac{\sqrt{2}}{4} \operatorname{Re} \sum_{ABn} \frac{\partial h_{AB}(\phi)}{\partial \phi_n} [- (\bar{\lambda}_B \gamma^\mu \gamma^\nu \psi_{nL}) f_{A\mu\nu} + 2i (\bar{\lambda}_B \psi_{nL}) D_A].
\end{aligned} \tag{27.4.42}$$

One interesting feature of this result is the appearance of a gaugino mass in theories with ϕ_n -dependent functions $h_{AB}(\phi)$, when supersymmetry is broken by a non-vanishing value of $\mathcal{F}_n$. This mechanism is used to generate gaugino masses in some theories of gravitationally mediated supersymmetry breaking, discussed in Section 31.7.

27.5 Supersymmetry Breaking in the Tree Approximation Resumed

We saw in the previous section that if the Fayet–Iliopoulos constants ξ_A all vanish and if there exists a solution of the equations $\partial f(\phi)/\partial \phi_n = 0$, then there is also some solution of these equations where the D -components of the gauge superfields all vanish, so that supersymmetry is unbroken. It follows that there are only two (non-exclusive) ways that supersymmetry can be spontaneously broken in the tree approximation in renormalizable theories of gauge and chiral superfields: the superpotential $f(\phi)$ may be arranged so that there are no solutions of all the equations $\partial f(\phi)/\partial \phi_n = 0$, or for gauge groups with $U(1)$ factors there may be Fayet–Iliopoulos terms in the action.

We have already seen in Section 26.5 how it can happen that there might not be any value of ϕ for which $\partial f(\phi)/\partial \phi_n = 0$. No change in that discussion is needed when the chiral

superfields interact with gauge fields, so let's turn to the other possibility: spontaneous supersymmetry breaking produced by Fayet–Iliopoulos terms. Since this can only arise for gauge groups with $U(1)$ factors, the simplest case is a theory with a single $U(1)$ gauge group. As discussed in Section 22.4, to avoid $U(1)$ - $U(1)$ - $U(1)$ and $U(1)$ -graviton-graviton anomalies it is necessary that the sum of the $U(1)$ quantum numbers of all left-chiral superfields and the sum of their cubes should vanish. We will consider the simplest possibility: two left-chiral superfields $\Phi_{\pm}$, with $U(1)$ quantum numbers $\pm e$. (This is a supersymmetric version of quantum electrodynamics, with the spinor components ψ_{-L} and ψ_{+L} of the two superfields providing the left-handed parts of the electron field and of its charge conjugate.) The most general $U(1)$ -invariant superpotential in a renormalizable theory is just $f(\Phi) = m\Phi_+\Phi_-$. The scalar potential (27.4.9) for the scalar components $\phi_{\pm}$ of these superfields is then

$$V(\phi_+, \phi_-) = m^2|\phi_+|^2 + m^2|\phi_-|^2 + (\xi + e^2|\phi_+|^2 - e^2|\phi_-|^2)^2. \quad (27.5.1)$$

Unless the Fayet–Iliopoulos constant ξ vanishes, it is evidently not possible to find a supersymmetric vacuum with $V = 0$. For $\xi > m^2/2e^2$ or $\xi < -m^2/2e^2$ the potential (27.5.1) has a minimum with either $\phi_+ = 0$ and $|\phi_-|^2 = (2e^2\xi - m^2)/2e^4$ or $\phi_- = 0$ and $|\phi_+|^2 = (-2e^2\xi - m^2)/2e^4$, so the $U(1)$ gauge symmetry is broken along with supersymmetry. For $|\xi| < m^2/2e^2$ the minimum of the potential is at $\phi_+ = \phi_- = 0$, so here the gauge symmetry is unbroken. There is in general no necessary connection between the possible breakdown of supersymmetry and of gauge symmetries.

Whether supersymmetry is spontaneously broken by the Fayet–Iliopoulos mechanism discussed here or by the O’Raifeartaigh mechanism of Section 26.5 or by some combination of the two, supersymmetry leaves a remnant in the pattern of tree-approximation masses. Inspection of the Lagrangian (27.4.8) for a general renormalizable supersymmetric theory of gauge and chiral superfields shows that the spontaneous breakdown of supersymmetry in this theory produces the following corrections to the masses calculated in Section 27.4.

Spin 0 Masses

If the $\mathcal{F}$ -terms $\mathcal{F}_n = -(\partial f(\phi)/\partial\phi_n)^*$ do not vanish at the minimum ϕ_0 of the potential, then the terms in the potential of second order in $\varphi_n \equiv \phi_n - \phi_{n0}$ have additional terms beyond those listed in Eq. (27.4.15):

$$\begin{aligned} V_{\text{quad}}(\varphi) = & \sum_{nm} (\mathcal{M}^* \mathcal{M})_{nm} \varphi_n^* \varphi_m + \sum_{Anm} (t_A \phi_0)_n (t_A \phi_0)_m^* \varphi_n^* \varphi_m \\ & + \frac{1}{2} \sum_{Anm} (t_A \phi_0)_n^* (t_A \phi_0)_m^* \varphi_n \varphi_m + \frac{1}{2} \sum_{Anm} (t_A \phi_0)_n (t_A \phi_0)_m \varphi_n^* \varphi_m^* \\ & + \frac{1}{2} \sum_{nm} \mathcal{N}_{nm} \varphi_n \varphi_m + \frac{1}{2} \sum_{nm} \mathcal{N}_{nm}^* \varphi_n^* \varphi_m^* + \sum_{Anm} D_{A0} (t_A)_{nm} \varphi_n^* \varphi_m, \end{aligned} \quad (27.5.2)$$

where $\mathcal{M}$ is again the complex symmetric matrix (??):

$$\mathcal{M}_{nm} \equiv \left(\frac{\partial^2 f(\phi)}{\partial\phi_n \partial\phi_m} \right)_{\phi=\phi_0},$$

$\mathcal{N}_{nm}$ is a new ingredient

$$\mathcal{N}_{nm} \equiv - \sum_{\ell} \mathcal{F}_{\ell 0} \left(\frac{\partial^3 f(\phi)}{\partial \phi_n \partial \phi_m \partial \phi_{\ell}} \right)_{\phi=\phi_0}, \quad (27.5.3)$$

and $\mathcal{F}_0$ and D_{A0} are again the $\mathcal{F}$ -terms and D -terms of the chiral scalar and gauge superfields at the minimum of the potential:

$$\mathcal{F}_{n0} = - \left[\frac{\partial f(\phi)}{\partial \phi_n} \right]_{\phi=\phi_0}^*, \quad D_{A0} = \xi_A + \sum_{nm} \phi_{n0}^* (t_A)_{nm} \phi_{m0}.$$

If we write the quadratic part (27.5.2) of the potential in the form (27.4.16):

$$V_{\text{quad}} = \frac{1}{2} \begin{bmatrix} \varphi \\ \varphi^* \end{bmatrix}^{\dagger} M_0^2 \begin{bmatrix} \varphi \\ \varphi^* \end{bmatrix},$$

then, instead of Eq. (27.4.17), we now have the scalar mass matrix

$$M_0^2 = \begin{bmatrix} \mathcal{M}^* \mathcal{M} + \mathcal{A} + \sum_A D_{A0} t_A & \mathcal{B} + \mathcal{N}^* \\ \mathcal{B}^* + \mathcal{N} & \mathcal{M} \mathcal{M}^* + \mathcal{A}^* + \sum_A D_{A0} t_A^{\text{T}} \end{bmatrix}, \quad (27.5.4)$$

where

$$\mathcal{A} \equiv \sum_A (t_A \phi_0) (t_A \phi_0)^{\dagger}, \quad \mathcal{B} \equiv \sum_A (t_A \phi_0) (t_A \phi_0)^{\text{T}}.$$

Spin 1/2 Masses

The fermion mass matrix M here is given again by Eq. (27.4.30):

$$M_{nm} = \mathcal{M}_{nm}, \quad M_{nA} = M_{An} = i\sqrt{2} (t_A \phi_0)_n^*, \quad M_{AB} = 0.$$

However, now instead of Eq. (27.4.19), the gauge invariance condition (27.4.18) yields

$$\sum_m \mathcal{M}_{nm} (t_A \phi_0)_m = \sum_m \mathcal{F}_{m0}^* (t_A)_{mn}. \quad (27.5.5)$$

Thus the Hermitian positive matrix whose eigenvalues are the squared masses of the fermions is given by

$$\begin{aligned} (M^{\dagger} M)_{nm} &= (\mathcal{M}^{\dagger} \mathcal{M})_{nm} + 2 \sum_A (t_A \phi_0)_n (t_A \phi_0)_m^*, \\ (M^{\dagger} M)_{AB} &= 2 (\phi_0^{\dagger} t_B t_A \phi_0), \\ (M^{\dagger} M)_{An} &= (M^{\dagger} M)_{nA}^* = i\sqrt{2} \sum_m \mathcal{F}_{m0} (t_A)_{mn}. \end{aligned} \quad (27.5.6)$$

Spin 1 Masses

The squared masses of the vector bosons are again given by the eigenvalues of the matrix (27.4.36):

$$(\mu^2)_{AB} = \left(\phi_0^\dagger \{t_B, t_A\} \phi_0 \right). \quad (27.5.7)$$

With the exception of the D -terms in Eq. (27.5.4), all of the changes in the mass-squared matrices are in their off-diagonal components. Therefore Eqs. (27.5.4), (27.5.6), and (27.5.7) yield particularly simple results for the traces of these matrices: for spin 0

$$\text{Tr } M_0^2 = 2 \text{Tr}(\mathcal{M}^* \mathcal{M}) + \text{Tr } \mu^2 + 2 \sum_A D_{A0} \text{Tr } t_A, \quad (27.5.8)$$

and for spin 1/2

$$\text{Tr}(M^\dagger M) = \text{Tr}(\mathcal{M}^* \mathcal{M}) + 2 \text{Tr } \mu^2. \quad (27.5.9)$$

Since the trace is the sum of the eigenvalues, we obtain from this a *mass sum rule*:

$$\sum_{\text{spin } 0} \text{mass}^2 - 2 \sum_{\text{spin } 1/2} \text{mass}^2 + 3 \sum_{\text{spin } 1} \text{mass}^2 = -2 \sum_A D_{A0} \text{Tr } t_A. \quad (27.5.10)$$

The trace of t_A automatically vanishes unless t_A is a $U(1)$ generator, and, as mentioned in Section 22.4, the trace must also vanish (when taken over all left-handed fermions) for $U(1)$ gauge generators to avoid gravitational contributions to an anomaly that would violate conservation of the $U(1)$ current. Thus (27.5.10) leads to the simpler result [4]

$$\sum_{\text{spin } 0} \text{mass}^2 - 2 \sum_{\text{spin } 1/2} \text{mass}^2 + 3 \sum_{\text{spin } 1} \text{mass}^2 = 0. \quad (27.5.11)$$

Of course, the unbroken conservation of charge, color, and baryon and lepton numbers prevents the mass matrices from having elements linking particles with different values of these quantum numbers, so all these results hold separately for each set of conserved quantum numbers.

The sum rule (27.5.11) is often quoted as providing evidence against models in which supersymmetry is spontaneously broken in the tree approximation within the minimum supersymmetric extension of the standard model. We will discuss this along with other arguments in Section 28.3.

As already observed in Section 26.5 (and discussed in greater generality in Sections 29.1 and 29.2), the spontaneous breakdown of supersymmetry necessarily entails the existence of a massless fermion, the goldstino. For renormalizable gauge theories in the tree approximation, the goldstino field g appears as a term in the spinor components ψ_n and λ_A of the chiral and gauge superfields, with coefficients given by

$$\psi_{nL} = i\sqrt{2}\mathcal{F}_{n0}g_L + \cdots, \quad \lambda_{AL} = D_{A0}g_L + \cdots, \quad (27.5.12)$$

where the dots denote terms involving spinor fields of definite non-zero mass. To check this we have to confirm that $(i\sqrt{2}\mathcal{F}_{n0}, D_{A0})$ is an eigenvector of the fermion mass-squared

matrix $M^\dagger M$ with eigenvalue zero. For this purpose we will need to use the condition that the potential (27.4.9) is stationary at $\phi = \phi_0$:

$$0 = \left. \frac{\partial V}{\partial \phi_n} \right|_{\phi=\phi_0} = - \sum_m \mathcal{M}_{nm} \mathcal{F}_{m0} + \sum_A D_{A0} (\phi_0^\dagger t_A)_n. \quad (27.5.13)$$

We also need the gauge invariance condition (27.4.12), which at $\phi = \phi_0$ reads

$$\sum_n \mathcal{F}_{n0} (t_A \phi_0)_n = 0. \quad (27.5.14)$$

Combining Eqs. (27.5.13) and (27.5.14) with Eqs. (27.5.5) and (27.5.6) then yields

$$\begin{aligned} i\sqrt{2} \sum_m (M^\dagger M)_{nm} \mathcal{F}_{m0} &= i\sqrt{2} \sum_A D_A (t_A \mathcal{F}_0^*)_n \\ &= - \sum_A (M^\dagger M)_{nA} D_{A0} \end{aligned} \quad (27.5.15)$$

and

$$\begin{aligned} i\sqrt{2} \sum_m (M^\dagger M)_{Am} \mathcal{F}_{m0} &= -2 \sum_{nm} \mathcal{F}_{n0} (t_A)_{nm} \mathcal{F}_{m0} \\ &= - \sum_B (M^\dagger M)_{AB} D_{B0}. \end{aligned} \quad (27.5.16)$$

That is,

$$M^\dagger M \begin{pmatrix} i\sqrt{2} \mathcal{F}_0 \\ D_0 \end{pmatrix} = 0, \quad (27.5.17)$$

as was to be shown.

27.6 Perturbative Non-Renormalization Theorems

From the beginning, several of the ultraviolet divergences in ordinary renormalizable quantum field theories were found to be absent in the supersymmetric versions of these theories. With the development in 1975 of supergraph techniques, in which all particles in each supermultiplet are considered together, it became possible to show that some radiative corrections are not only finite, but are absent altogether in perturbation theory [5]. Supergraphs will be described in detail in Chapter 30, but as it happens they are not needed to prove the most important non-renormalization theorems. This section will give a version of a method developed by Seiberg [6] in 1993, which showed how the non-renormalization theorems may be easily obtained from simple considerations of symmetry and analyticity.

Consider a general renormalizable supersymmetric gauge theory with a number of left-chiral superfields Φ_n and/or gauge superfields V_A . As mentioned in Section 27.3, if we remove the factor g in the t_A and C_{ABC} and include it instead in the gauge superfields, then the Lagrangian density will be of the form

$$\mathcal{L} = \left[\Phi^\dagger e^{-V} \Phi \right]_D + 2 \operatorname{Re} [f(\Phi)]_{\mathcal{F}} + \frac{1}{2g^2} \operatorname{Re} \left[\sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}, \quad (27.6.1)$$

where the superpotential $f(\Phi)$ is a gauge-invariant cubic polynomial in the left-chiral superfields. (We are ignoring a possible θ -term, which has no effect in perturbation theory.)

Suppose we impose an ultraviolet cut-off λ on the momenta circulating in loop graphs. As discussed in Section 12.4, we can find a *local* ‘Wilsonian’ effective Lagrangian density $\mathcal{L}_\lambda$ that, with this cut-off, gives precisely the same results as the original Lagrangian density for S -matrix elements of processes at momenta below λ . The effective Lagrangian density has masses and coupling parameters that now depend on λ , and usually there will be an infinite number of coupling terms in the effective Lagrangian density, all possible terms allowed by the symmetries of the theory. But things are much simpler in supersymmetric theories. The non-renormalization theorems tell us that, as long as the cut-off preserves supersymmetry and gauge invariance, to all orders in perturbation theory the effective Lagrangian will have the structure

$$\begin{aligned} \mathcal{L}_\lambda = & \left[\mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, \mathcal{D} \dots) \right]_D + 2 \operatorname{Re} [f(\Phi)]_{\mathcal{F}} \\ & + \frac{1}{2g_\lambda^2} \operatorname{Re} \left[\sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}, \end{aligned} \quad (27.6.2)$$

where $\mathcal{A}_\lambda$ is a general Lorentz- and gauge-invariant function; ‘ $\mathcal{D} \dots$ ’ denotes terms involving superderivatives or spacetime derivatives of the preceding arguments; and g_λ is the *one-loop* effective gauge coupling, given by the same formula as the one-loop renormalized gauge coupling constant

$$g_\lambda^{-2} = \text{constant} - 2b \ln \lambda, \quad (27.6.3)$$

where b is the coefficient of g^3 in the Gell-Mann–Low function $\beta(g)$, discussed in Chapter 18. This is for a simple gauge group, with a single gauge coupling, but the extension to a direct product of simple and $U(1)$ gauge groups is trivial. Note in particular that the effective superpotential is not only finite in the limit $\lambda \rightarrow \infty$, but at least in perturbation theory it contains no terms beyond those in the original superpotential, and there is no change in the coefficients of the terms it does contain.

To prove this theorem, we shall interpret this theory as a special case of one with two additional external gauge-invariant left-chiral superfields X and Y , with Lagrangian density

$$\mathcal{L}^\# = \frac{1}{2} \left[\Phi^\dagger e^{-V} \Phi \right]_D + 2 \operatorname{Re} [Y f(\Phi)]_{\mathcal{F}} + \frac{1}{2} \operatorname{Re} \left[X \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}. \quad (27.6.4)$$

This Lagrangian density becomes equal to the original one when the scalar components x and y of X and Y are given the values $x = 1/g^2$ and $y = 1$, and the spinor and auxiliary components of X and Y are set equal to zero. Since supersymmetry and gauge invariance are assumed to be preserved in the cut-off procedure, the effective Lagrangian density in the presence of these external superfields must be the sum of the D -term of a general superfield and the real part of the $\mathcal{F}$ -term of a left-chiral superfield:

$$\begin{aligned} \mathcal{L}_\lambda^\# = & \left[\mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, X, X^\dagger, Y, Y^\dagger, \mathcal{D} \dots) \right]_D \\ & + 2 \operatorname{Re} [\mathcal{B}_\lambda(\Phi, W_L, X, Y)]_{\mathcal{F}}, \end{aligned} \quad (27.6.5)$$

with $\mathcal{A}_\lambda$ and $\mathcal{B}_\lambda$ both gauge-invariant functions of the displayed arguments. We do not include any superderivatives or spacetime derivatives in the $\mathcal{F}$ -term because, as in Section 26.3, terms involving derivatives of any of the left-chiral superfields or their adjoints may be rewritten as contributions to $[\mathcal{A}_\lambda]_D$. (It is true that Eq. (27.3.12) gives W_L itself in terms of two $\mathcal{D}_{RS}$ acting on a superfield $\exp(-2V)\mathcal{D}_L\exp(2V)$, but this superfield is not gauge-invariant, and we are requiring that $\mathcal{A}_\lambda$ be gauge-invariant.)

The dependence of $\mathcal{B}_\lambda$ on X and Y is severely limited by two additional symmetries of the action obtained from the Lagrangian density (27.6.4). (Both of these symmetries are broken by non-perturbative effects, which will be considered in Chapter 29.) The first symmetry is a perturbative $U(1)$ R -symmetry, of the sort discussed in Section 26.3, for which θ_L and θ_R are given the R values $+1$ and -1 , the superfields Φ , V , and X are R -neutral, and Y has the R value $+2$. (Recall that $f_{\mathcal{F}}$ is the coefficient of θ_L^2 in f , so in order for $f_{\mathcal{F}}$ to have the R value 0, f must have the R value 2.) Because W_L is given by two $\mathcal{D}_{RS}$ and one $\mathcal{D}_L$ acting on R -neutral superfields, it has the R value $+1$. Now, R invariance requires $\mathcal{B}_\lambda$, like the superpotential, to have the R value $+2$. It cannot depend on any superfields with negative R values, such as adjoints of left-chiral superfields, because it is *holomorphic*, so the terms in $\mathcal{B}_\lambda$ can only be of first order in Y or of second order in W_L , with coefficients depending only on the R -neutral superfields Φ and/or X :

$$\begin{aligned} \mathcal{B}_\lambda(\Phi, W_L, X, Y) &= Y f_\lambda(\Phi, X) \\ &+ \sum_{\alpha\beta AB} \epsilon_{\alpha\beta} W_{A\alpha L} W_{B\beta L} h_{\lambda AB}(\Phi, X). \end{aligned} \quad (27.6.6)$$

(Lorentz invariance requires the spinor indices on the W_L s to be contracted with $\epsilon_{\alpha\beta}$.) The other symmetry is translation of X by an imaginary numerical constant, $X \rightarrow X + i\xi$, with ξ real. This changes the Lagrangian density (27.6.4) by an amount proportional to $\text{Im} \sum_{A\alpha\beta} W_{A\alpha L} W_{A\beta L}$, which as we saw in Section 27.3 is a spacetime derivative, and therefore can have no effect in perturbation theory. This translation symmetry prevents X from appearing anywhere in the effective Lagrangian density (27.6.5) except where it appears in the original Lagrangian density (27.6.4). We therefore conclude that f_λ is independent of X , while $h_{\lambda AB}$ consists of a Φ -independent term proportional to $X\delta_{AB}$, plus a term that is independent of X . That is,

$$\begin{aligned} \mathcal{B}_\lambda(\Phi, W_L, X, Y) &= Y f_\lambda(\Phi) \\ &+ \sum_{\alpha\beta AB} \epsilon_{\alpha\beta} W_{A\alpha L} W_{B\beta L} [c_\lambda \delta_{AB} X + \ell_{\lambda AB}(\Phi)], \end{aligned} \quad (27.6.7)$$

where c_λ is a real cut-off-dependent constant.

The point of introducing the external auxiliary superfields X and Y is that, by giving them suitable values, we can make use of weak-coupling approximations to determine the coefficients in Eq. (27.6.7). If we set the spinor and auxiliary components of X and Y equal to zero, and take their scalar components x and y to approach infinity and zero, respectively, then the gauge coupling constant vanishes as $1/\sqrt{x}$, and all Yukawa and scalar couplings derived from the superpotential vanish as y . In this limit, the only graph that

contributes to the term in (27.6.7) proportional to Y has a single vertex arising from the term $2 \operatorname{Re}[Y f(\Phi)]_{\mathcal{F}}$ in Eq. (27.6.4), so

$$f_{\lambda}(\Phi) = f(\Phi). \quad (27.6.8)$$

Also, with $Y = 0$ there is a conservation law which requires every term in $\mathcal{L}_{\lambda}^{\#}$ to have equal numbers of Φ s and $\Phi^{\dagger}$ s, so since $\Phi^{\dagger}$ cannot occur in $\ell_{\lambda AB}$, neither can Φ . Gauge invariance then requires the constant $\ell_{\lambda AB}$ to be proportional to δ_{AB} for a simple group:

$$\ell_{\lambda AB} = \delta_{AB} L_{\lambda}. \quad (27.6.9)$$

Now, since gauge propagators go as $1/x$ while pure gauge interactions go as x and scalar propagators and interactions are x -independent, with $y = 0$ the number of powers of x in a diagram with V_W pure gauge boson vertices, I_W internal gauge boson lines, and any number of scalar-gauge boson vertices and scalar propagators is

$$N_x = V_W - I_W. \quad (27.6.10)$$

The number of loops is given by

$$L = I_W + I_{\Phi} - V_W - V_{\Phi} + 1, \quad (27.6.11)$$

where I_{Φ} is the number of internal Φ lines and V_{Φ} is the number of Φ - V interaction vertices. All the Φ - V vertices have two Φ lines attached, so with no external Φ lines I_{Φ} and V_{Φ} are equal, and therefore cancel in Eq. (27.6.11), so that Eq. (27.6.10) may be written

$$N_x = 1 - L. \quad (27.6.12)$$

Thus the coefficient c_{λ} of X in Eq. (27.6.7) is correctly given by the tree approximation and is therefore what it was in the original Lagrangian, simply $c_{\lambda} = 1$, while the coefficient L_{λ} of the X -independent term is given by one-loop diagrams only. Putting this all together, we have

$$\begin{aligned} \mathcal{L}_{\lambda}^{\#} = & \left[\mathcal{A}_{\lambda}(\Phi, \Phi^{\dagger}, V, X, X^{\dagger}, Y, Y^{\dagger}, \mathcal{D} \cdots) \right]_D + 2 \operatorname{Re}[Y f(\Phi)]_{\mathcal{F}} \\ & + \frac{1}{2} \operatorname{Re} \left[(X + L_{\lambda}) \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}, \end{aligned} \quad (27.6.13)$$

where L_{λ} is the one-loop contribution. Setting $Y = 1$ and $X = 1/g^2$ then gives Eq. (27.6.2), with $g_{\lambda}^{-2} = g^{-2} + L_{\lambda}$. As shown in Section 18.3, the leading order contribution to $\lambda dg_{\lambda}/d\lambda$ is the same function of g_{λ} whatever renormalization scheme is used to define this coupling, so to one-loop order we must have

$$\lambda \frac{dg_{\lambda}}{d\lambda} = b g_{\lambda}^3, \quad (27.6.14)$$

where b is the same coefficient of g^3 as in the renormalization group equation of Gell-Mann and Low. The solution is Eq. (27.6.3), completing the proof.

In theories with a $U(1)$ gauge superfield V_1 , the Lagrangian may contain a Fayet–Iliopoulos term (27.2.7):

$$\mathcal{L}_{\text{FI}} = \xi[V_1]_D. \quad (27.6.15)$$

It is easy to see that the coefficient ξ of such a term is not renormalized [7]. If the corresponding coefficient ξ_λ in the Wilsonian Lagrangian density did depend on the gauge couplings or the couplings in the superpotential, then when we replace the original Lagrangian (27.6.1) with a Lagrangian (27.6.4) involving the external superfields X and Y , this term in the Wilsonian Lagrangian would be required by supersymmetry to take the form

$$\mathcal{L}_{\text{FI}\lambda}^\# = [\xi_\lambda(X, Y, X^*, Y^*)V_1]_D, \quad (27.6.16)$$

with ξ_λ a function with a non-trivial dependence on X and/or Y and/or their adjoints. But such a term would not be gauge-invariant, because, according to Eq. (27.2.18), a gauge transformation shifts V_1 by a chiral superfield $i(\Omega - \Omega^*)/2$, and although the D -term of a chiral superfield vanishes, the product of $i(\Omega - \Omega^*)/2$ and ξ_λ is not chiral for general gauge transformations if ξ_λ has any dependence on other superfields. There actually are diagrams that make contributions to ξ_λ that are independent

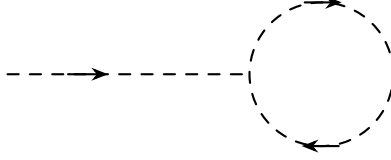


Figure 27.1. A one-loop diagram that could be quadratically divergent in theories with supersymmetry broken by trilinear couplings among scalar fields and their adjoints. The lines all represent complex scalar fields.

of all coupling constants. For the Lagrangian (27.6.1) there are no factors of the gauge coupling g at vertices at which the gauge superfield interacts with chiral matter, but instead a factor g^{-2} for each gauge propagator, so a graph with no internal gauge lines and no self-couplings of the chiral superfield will have no dependence on coupling constants. The only such graphs that contribute to ξ_λ are those in which a single external gauge line is attached to a chiral loop. (See Figure 27.1.) The contribution of all such graphs is proportional to the sum of the gauge couplings of all chiral superfields—that is, to the trace of the $U(1)$ generator. But as discussed in Section 22.4, this trace must vanish (if the $U(1)$ symmetry is unbroken) in order to avoid gravitational anomalies that violate the conservation of the $U(1)$ current.

The most important application of these theorems is a corollary, which tells us that if there is no Fayet–Iliopoulos term and if the superpotential $f(\Phi)$ allows solutions of the equations $\partial f(\phi)/\partial\phi_n = 0$, then supersymmetry is not broken in any finite order of perturbation theory.

To test this we must examine Lorentz-invariant field configurations, in which the Φ_n have only constant scalar components ϕ_n and constant auxiliary components $\mathcal{F}_n$, while (in Wess–Zumino gauge) the coefficients V_A of the gauge generators t_A in the matrix gauge

superfield V have only auxiliary components D_A . Supersymmetry is unbroken if there are values of ϕ_n for which $\mathcal{L}_\lambda$ has no terms of first order in $\mathcal{F}_n$ or D_A , in which case there is sure to be an equilibrium solution with $\mathcal{F}_n = D_A = 0$. (In Section 29.2 we will see that this is the sufficient as well as the necessary condition for supersymmetry to be unbroken.) In the absence of Fayet–Iliopoulos terms, this will be the case if for all A

$$\sum_{nm} \frac{\partial K_\lambda(\phi, \phi^*)}{\partial \phi_n^*} (t_A)_{mn} \phi_m^* = 0 \quad (27.6.17)$$

and for all n

$$\frac{\partial f(\phi)}{\partial \phi_n} = 0, \quad (27.6.18)$$

where the effective Kähler potential $K_\lambda(\phi, \phi^*)$ is

$$K_\lambda(\phi, \phi^*) = \mathcal{A}_\lambda(\phi, \phi^*, 0, 0 \cdots), \quad (27.6.19)$$

with $\mathcal{A}_\lambda(\phi, \phi^*, 0, 0 \cdots)$ obtained from $\mathcal{A}_\lambda$ by setting the gauge superfield and all superderivatives equal to zero. (With superderivatives required to vanish by Lorentz invariance, the only dependence of $\mathcal{A}_\lambda$ on V is a factor $\exp(-V)$ following every factor $\Phi^\dagger$.)

We now use a trick that we have already employed in Section 27.4. If there is any solution $\phi^{(0)}$ of Eq. (27.6.18), then the gauge symmetry tells us that there is a continuum of such solutions, with ϕ_n replaced with

$$\phi_n(z) = \left[\exp \left(i \sum_A t_A z_A \right) \right]_{nm} \phi_m^{(0)}, \quad (27.6.20)$$

where (since f depends only on ϕ , not ϕ^*) the z_A are an arbitrary set of *complex* parameters. If $K_\lambda(\phi, \phi^*)$ has a stationary point anywhere on the surface $\phi = \phi(z)$, then at that point

$$\begin{aligned} 0 = & \sum_{nmA} \frac{\partial K_\lambda(\phi, \phi^*)}{\partial \phi_n} (t_A)_{nm} \phi_m \delta z_A \\ & - \sum_{nmA} \frac{\partial K_\lambda(\phi, \phi^*)}{\partial \phi_n^*} (t_A)_{mn} \phi_m^* \delta z_A^*. \end{aligned} \quad (27.6.21)$$

Since this must be satisfied for all infinitesimal complex δz_A , the coefficients of both δz_A and δz_A^* must both vanish, and therefore Eq. (27.6.17) as well as Eq. (27.6.18) is satisfied at this point. Thus the existence of a stationary point of $K_\lambda(\phi, \phi^*)$ on the surface $\phi = \phi(z)$ would imply that supersymmetry is unbroken to all orders of perturbation theory.

The zeroth-order Kähler potential $(\phi^\dagger \phi)$ is bounded below and goes to infinity as $\phi \rightarrow \infty$, so it certainly has a minimum on the surface $\phi = \phi(z)$, where of course it is stationary. If there were no flat directions in which K_λ is constant at this minimum then any sufficiently small perturbation to the Kähler potential might shift the minimum, but would not destroy it. At the minimum of the Kähler potential on the surface $\phi = \phi(z)$ there are flat directions: ordinary global gauge transformations $\delta \phi = i \sum_A \delta z_A t_A \phi$ with z_A real. But these are also flat directions for the perturbation $K_\lambda(\phi, \phi^*) - (\phi^\dagger \phi)$, so there is still a local minimum of

K_λ on the surface $\phi = \phi(z)$ for any perturbation in at least a finite range, and thus to all orders in whatever couplings appear in $K_\lambda(\phi, \phi^*)$. As we have seen, this is a set of scalar field values at which $\mathcal{F}_n = 0$ and $D_A = 0$ for all n and A , which means that supersymmetry is unbroken.

* * *

These results may be extended to non-renormalizable theories [3]. In such theories the first term $[\Phi^\dagger e^{-V} \Phi]_D$ in Eq. (27.6.1) is replaced with the D -term of an arbitrary real gauge-invariant scalar function of $\Phi^\dagger$, Φ , V , and their superderivatives and spacetime derivatives, while the second and third terms in Eq. (27.6.1) are replaced with the $\mathcal{F}$ -term of an arbitrary globally gauge-invariant scalar function $f(\Phi, W)$ of Φ_n and W_α . It has been shown that, to all orders of perturbation theory, the function $f_\lambda(\Phi, W)$ appearing in the $\mathcal{F}$ -term of the Wilsonian Lagrangian is the same as $f(\Phi, W)$, except for the one-loop renormalization of the term quadratic in W .

27.7 Soft Supersymmetry Breaking⁴

We will see in the next chapter that, even if supersymmetry is an exact symmetry of the action, the spontaneous breakdown of supersymmetry at very high energy can produce superrenormalizable terms that violate supersymmetry conservation in the effective action that describes physics at lower energies. These superrenormalizable terms may explain the lack of supersymmetry observed in phenomena at accessible energies. In this section we will consider the radiative corrections that can be produced by such supersymmetry-breaking superrenormalizable terms, in part to see whether this provides a criterion for including or rejecting such terms in supersymmetric versions of the standard model.

The sign of supersymmetry breaking is the appearance of expectation values of D -terms of general superfields or $\mathcal{F}$ -terms of chiral superfields. Any operator $\epsilon\mathcal{O}$ in the Lagrangian density that breaks supersymmetry can be written in a supersymmetric form, as a D -term

$$\epsilon\mathcal{O} = [ZS]_D, \quad (27.7.1)$$

where S is a non-chiral superfield that has $\mathcal{O}$ as its C -term, and Z is a non-chiral external superfield whose only non-vanishing component is $[Z]_D = \epsilon$. Some but not all operators $\epsilon\mathcal{O}$ that break supersymmetry can also be written as $\mathcal{F}$ -terms,

$$\epsilon\mathcal{O} = [\Omega\mathcal{O}]_{\mathcal{F}}, \quad (27.7.2)$$

or their adjoints, where $\mathcal{O}$ is the left-chiral superfield whose $\mathcal{F}$ -term is $\mathcal{O}$ and Ω is an external left-chiral superfield whose only non-vanishing component is $[\Omega]_{\mathcal{F}} = \epsilon$. We can count the order in ϵ in which a given correction to the effective Lagrangian will occur by counting the powers of Z or Ω needed to construct this correction in a supersymmetric way. We will find interesting limitations on the radiative corrections that can be produced by those interactions that can be written in the form (27.7.2) as well as (27.7.1).

⁴This section lies somewhat out of the book's main line of development and may be omitted in a first reading.

According to the results of the previous section, there are no radiative corrections to $\mathcal{F}$ -terms, so all supersymmetry-breaking radiative corrections to the Wilsonian Lagrangian density must take the form of D -terms. This theorem does not prevent any given operator from appearing in the Wilsonian Lagrangian density, because even if an operator $\epsilon\Delta\mathcal{L}$ cannot be expressed in the form $[Z\Lambda]_D$, where Λ is the general superfield whose C -term is $\Delta\mathcal{L}$, yet $\epsilon^2\Delta\mathcal{L}$ can be expressed in the form

$$\epsilon^2\Delta\mathcal{L} = 2[\Omega^*\Omega\Lambda]_D. \quad (27.7.3)$$

But not all operators can be produced by radiative corrections of *first order* in Ω or Ω^* . In particular, a function only of the ϕ -term of a left-chiral superfield Φ , but not of ϕ^* , cannot be written as the D -term of a superfield linear in Ω . (Note that $[\Omega h(\Phi)]_D$ is a derivative, while $[\Omega^* h(\Phi)]_D = 2[\Phi]_{\mathcal{F}} \partial h(\phi)/\partial\phi$ is not a function of ϕ alone.) We conclude then that *the supersymmetry-breaking terms in the Wilsonian Lagrangian that depend on ϕ alone cannot be produced by radiative corrections that are of first order in supersymmetry-breaking interactions of the form (27.7.2).*

This result is significant because the most divergent radiative corrections are those that are of lowest order in superrenormalizable couplings. To be more specific, the coefficient of an interaction of dimensionality $\mathcal{D}$ has dimensionality (in powers of energy) $4 - \mathcal{D}$, so dimensional analysis indicates that the contribution of a set of interactions of dimensionality d_1, d_2 , etc. to the coefficient of an interaction of dimensionality d can contain the ultraviolet cut-off to at most a power

$$p = 4 - d - (4 - d_1) - (4 - d_2) - \dots \quad (27.7.4)$$

and is therefore finite if $p < 0$. (This argument ignores possible ultraviolet divergences in subintegrations; for a thorough treatment of this topic, see Reference 8.) Superrenormalizable interactions are ‘soft,’ in the sense that they reduce the degree of divergence of the graphs in which they appear. In particular, in a renormalizable theory where all interactions have $d_i \leq 4$, and the strictly renormalizable interactions with $d_i = 4$ are supersymmetric, the contribution of one or more superrenormalizable interactions to the coefficient of an interaction with $d = 4$ will always have $p < 0$, so even if they are not supersymmetric the superrenormalizable interactions will not produce supersymmetry-violating ultraviolet-divergent corrections to the coefficients of the supersymmetric $d = 4$ interactions.

On the other hand, in such a theory there may be divergent radiative corrections to the superrenormalizable interactions themselves [9]. The most worrisome are quadratic (or higher) divergences, which if cut off at some high energy scale M_X may require a fine-tuning of the bare coupling constants to preserve supersymmetry as a good approximate symmetry at energies below M_X . According to Eq. (27.7.4), in renormalizable theories where all interactions with $d_i = 4$ are supersymmetric, radiative corrections can produce quadratic or more highly divergent ($p \geq 2$) supersymmetry-violating operators of dimensionality d only if they involve insertion of a superrenormalizable supersymmetry-violating interaction of dimensionality $d_1 \geq 2 + d$. This allows either $d = 0$ and $d_1 \geq 2$, which arises only when we calculate the cosmological constant, or $d = 1$ and $d_1 = 3$, which arises only when

we calculate the ‘tadpole’ graphs in which a scalar field line disappears into the vacuum. The cosmological constant raises fine-tuning problems for all known theories [10] and will not be considered further here. The tadpole graphs represent operators linear in ϕ or ϕ^* and, as we have seen, can not be produced to first order in supersymmetry-violating interactions that can be put in the form (27.7.2). Such superrenormalizable interactions are therefore ‘soft’ in the sense that they do not induce quadratic or higher divergences. Along with the superrenormalizable interactions with $d \leq 2$, including arbitrary quadratic polynomials in ϕ and ϕ^* , the supersymmetry-breaking interactions that are soft in this sense include terms of third order in the ϕ , which can be expressed as $\phi^3 = [\Omega\Phi^3]_{\mathcal{F}}$, and likewise terms of third order in the ϕ^* , and also $d = 3$ gaugino mass terms, which can be written as $[\Omega\epsilon_{\alpha\beta}W_\alpha W_\beta]_{\mathcal{F}}$, but not terms like $\phi^2\phi^*$ or $\phi\phi^{2*}$, which in general *can* produce quadratically divergent tadpole graphs [9].

Nevertheless, tadpoles can only arise for scalar fields that are neutral with respect to all exact symmetries. In theories without such neutral scalars, like the supersymmetric standard model discussed in the next chapter, *all* superrenormalizable interactions may be considered to be soft.

27.8 Another Approach: Gauge-Invariant Supersymmetry Transformations⁵

It is somewhat disturbing that the supersymmetry transformation rules discussed so far involve ordinary spacetime derivatives, not gauge-invariant derivatives. For instance, in a $U(1)$ gauge theory, the transformation of the component fields of a chiral scalar superfield is given by Eqs. (??)–(??) as

$$\begin{aligned}\delta\psi_L &= \sqrt{2}\partial_\mu\phi\gamma^\mu\alpha_R\phi + \sqrt{2}\mathcal{F}\alpha_L, \\ \delta\mathcal{F} &= \sqrt{2}(\bar{\alpha}_L\not{\partial}\psi_L), \\ \delta\phi &= \sqrt{2}(\bar{\alpha}_R\psi_L).\end{aligned}\tag{27.8.1}$$

One might have thought that in the transformation of a chiral superfield that carries $U(1)$ charge q , the ordinary spacetime derivatives in Eq. (27.8.1) should be replaced with gauge-covariant derivatives, given in terms of the $U(1)$ gauge field V_μ by

$$D_\mu = \partial_\mu - iqV_\mu.\tag{27.8.2}$$

With such gauge-invariant supersymmetry transformations of the chiral superfields, one would still attempt to formulate supersymmetry transformations of the gauge supermultiplet that involve only the physical and auxiliary fields V_μ , λ , and D :

$$\begin{aligned}\tilde{\delta}V_\mu &= (\bar{\alpha}\gamma_\mu\lambda), \\ \tilde{\delta}\lambda &= iD\gamma_5\alpha + \frac{1}{2}[\partial_\mu V, \gamma^\mu]\alpha, \\ \tilde{\delta}D &= i(\bar{\alpha}\gamma_5\not{\partial}\lambda),\end{aligned}\tag{27.8.3}$$

⁵This section lies somewhat out of the book’s main line of development and may be omitted in a first reading.

in which ordinary spacetime derivatives appear because the gauge superfield carries no $U(1)$ charge.

This doesn't work. The algebra of these transformations does not close: the commutator of two of the modified supersymmetry transformations is not a linear combination of bosonic symmetry transformations, such as spacetime translations and gauge transformations. It follows that it is not possible to construct a Lagrangian for the chiral and gauge superfields that would be invariant under these modified supersymmetry transformations, because if there were such a Lagrangian then it would have to be also invariant under the commutators of these transformations, so that these commutators would have to be bosonic symmetries of the Lagrangian.

In 1973 de Wit and Freedman [11] showed that the supersymmetry algebra could be made to close by modifying the supersymmetry transformation properties of the chiral superfields not only by changing ordinary derivatives to gauge-invariant derivatives, but by also adding an extra term in the transformation of the $\mathcal{F}$ -component, so that for $U(1)$ gauge theories the modified supersymmetry transformation rules read

$$\begin{aligned}\tilde{\delta}\psi_L &= \sqrt{2} D_\mu \phi \gamma^\mu \alpha_R \phi + \sqrt{2} \mathcal{F} \alpha_L, \\ \tilde{\delta}\mathcal{F} &= \sqrt{2} (\bar{\alpha}_L \not{D} \psi_L) - 2iq\phi(\bar{\alpha}_L \lambda_R), \\ \tilde{\delta}\phi &= \sqrt{2} (\bar{\alpha}_R \psi_L).\end{aligned}\tag{27.8.4}$$

With this change, they were also able to construct a Lagrangian that is invariant under the transformations (27.8.3)–(27.8.4) and that turned out to be just the one we have found in Sections 27.1 and 27.2.

There is nothing wrong with continuing to use the conventional transformation rules (27.8.1), so we do not need the de Wit–Freedman formalism to deal with supersymmetric gauge theories. Nevertheless this formalism is of some interest, because in supergravity theories the analog of the conventional formalism is very cumbersome. As described in Chapter 31, the formalism that has been chiefly used to derive physically interesting results in supergravity theories follows an approach like that of de Wit and Freedman, with supersymmetry transformation rules involving covariant derivatives in place of ordinary derivatives, rather than an approach based on conventional supersymmetry transformations like those of Eq. (27.8.1). It is therefore of some interest to understand the relation between the de Wit–Freedman formalism and the conventional approach in the relatively simple context of $U(1)$ gauge theory, and in particular to explain the origin of the extra term in the transformation rule for $\mathcal{F}$.

In writing supersymmetry transformations (27.8.3) that did not involve the components C , M , N , or ω of the gauge superfield V , de Wit and Freedman were implicitly adopting the Wess–Zumino gauge discussed in Section 27.1. But the choice of Wess–Zumino gauge is not invariant under either the conventional supersymmetry transformations (??)–(??) or the extended gauge transformations (27.1.17), so once we adopt this gauge both symmetries are lost. We can, however, define a *combined* transformation, acting on fields in Wess–Zumino gauge, which consists of a conventional supersymmetry transformation followed by

an extended gauge transformation that takes us back to Wess–Zumino gauge. *This is the de Wit–Freedman transformation $\tilde{\delta}$* ⁶.

To construct the de Wit–Freedman transformations in this way, note that for a gauge superfield that satisfies the Wess–Zumino gauge conditions $C = M = N = \omega = 0$, the transformation rules (??)–(??) give

$$\delta C = 0, \quad \delta \omega = \not{V} \alpha, \quad \delta M = -(\bar{\alpha} \lambda), \quad \delta N = i(\bar{\alpha} \gamma_5 \lambda). \quad (27.8.5)$$

According to Eqs. (27.1.17), we can get back to Wess–Zumino gauge by performing an infinitesimal extended gauge transformation (27.1.13):

$$V \longrightarrow V + \frac{i}{2} [\Omega - \Omega^*], \quad (27.8.6)$$

where Ω is a left-chiral superfield with components

$$\phi^\Omega = 0, \quad \psi_L^\Omega = -\sqrt{2} \not{V} \alpha_R, \quad \mathcal{F}^\Omega = -(\bar{\alpha}(\mathbf{1} - \gamma_5) \lambda). \quad (27.8.7)$$

According to Eq. (27.1.11), this extended gauge transformation induces on a chiral superfield of charge q the transformation

$$\delta' \Phi = iq \Omega \Phi. \quad (27.8.8)$$

Using the multiplication rules (??)–(??), the transformation of the components of Φ is

$$\begin{aligned} \delta' \psi_L &= -i\sqrt{2} q \phi \not{V} \alpha_R, \\ \delta' \mathcal{F} &= -2iq \phi (\bar{\alpha}_L \lambda_R) - i\sqrt{2} q (\bar{\alpha}_L \not{V} \psi_L), \\ \delta' \phi &= 0. \end{aligned} \quad (27.8.9)$$

Adding this to Eq. (27.8.1) and comparing with Eq. (27.8.4) shows that the de Wit–Freedman transformation is indeed the combination of a conventional supersymmetry transformation and the corresponding extended gauge transformation (27.8.8):

$$\tilde{\delta} \Phi = \delta \Phi + \delta' \Phi. \quad (27.8.10)$$

27.9 Gauge Theories with Extended Supersymmetry⁷

Theories with unbroken extended supersymmetry are not considered to be good candidates for realistic extensions of the standard model, because of the non-chirality of particle multiplets discussed in Section 25.4. Nevertheless gauge theories with extended supersymmetry are worth some consideration here because they have provided paradigms for the use of powerful mathematical methods to solve dynamical problems.

⁶This was not shown explicitly by de Wit and Freedman. But, in fact, although the point of their paper was to emphasize that the details of the transformations (27.8.3) and (27.8.4) could be inferred from the requirement of a closed supersymmetry algebra (also for non-Abelian gauge theories), they remarked that they had actually found these transformations by identifying the fermionic transformations that survive in Wess–Zumino gauge.

⁷This section lies somewhat out of the book’s main line of development and may be omitted in a first reading.

There are a number of special formalisms that have been proposed to construct Lagrangians with $N = 2$ extended supersymmetry [12], but fortunately we can get by with the tools already at hand. Any theory with $N = 2$ supersymmetry also has $N = 1$ supersymmetry, so its Lagrangian must be a special case of the Lagrangians already considered in this chapter. To construct a Lagrangian with $N = 2$ supersymmetry for some set of the $N = 2$ supermultiplets of particles constructed in Sections 25.4 and 25.5, we need only write down the most general Lagrangian with $N = 1$ supersymmetry whose $N = 1$ supermultiplets contain physical fields for the particles in the $N = 2$ supermultiplets, and then impose a discrete R -symmetry on the Lagrangian: a symmetry that acts differently on different components of $N = 2$ supermultiplets. The Lagrangian density will then be invariant under a second supersymmetry, whose supermultiplets are given by acting on the supermultiplets of ordinary $N = 1$ supersymmetry with the R -symmetry.

It will be convenient to choose the discrete R -transformation so that

$$Q_1 \longrightarrow Q_2, \quad Q_2 \longrightarrow -Q_1. \quad (27.9.1)$$

If the central charge were zero then the supersymmetry algebra would be invariant under an $SU(2)$ R -symmetry group, which has the transformation (27.9.1) as one finite element $\exp(i\pi\tau_2/2)$, but symmetry under the discrete symmetry is sufficient for our purposes, so we do not need to assume a zero central charge. In fact, it will turn out that the Lagrangians we construct by this method will have an $SU(2)$ R -symmetry, not just symmetry under the discrete transformation (27.9.1).

Let us first consider the renormalizable theory of the gauge bosons of a general gauge group, together with the superpartners required by $N = 2$ extended supersymmetry. We saw in Section 25.4 that in $N = 2$ global supersymmetry theories a massless gauge boson can only belong to a multiplet also containing a pair of massless fermions of each helicity $\pm 1/2$ that transform as a doublet under the $SU(2)$ R -symmetry and a pair of $SU(2)$ -singlet spinless bosons. Since $N = 2$ supersymmetry includes $N = 1$ supersymmetry, the renormalizable Lagrangian for this theory must be a special case of the general renormalizable Lagrangian density (27.4.1). One feature of this special case is that since the gauge boson belongs to the adjoint representation of the gauge group, so must also the fermions and scalar field. To furnish $N = 2$ supermultiplets of fields with the correct particle content, we must have one $N = 1$ chiral superfield Φ_A with component fields $\phi_A, \psi_A, \mathcal{F}_A$ (with ψ_A Majorana and ϕ_A and $\mathcal{F}_A$ both complex) for each $N = 1$ gauge multiplet V_A^μ, λ_A, D_A . We impose a discrete R -symmetry under the transformation

$$\psi_A \longrightarrow \lambda_A, \quad \lambda_A \longrightarrow -\psi_A, \quad (27.9.2)$$

(with all other fields invariant) because this is the effect of the transformation (27.9.1). Since a non-trivial superpotential would give ψ_A interactions or mass terms that are absent for λ_A , the superpotential must vanish. The Lagrangian (27.4.1) therefore takes the special

form

$$\begin{aligned}
\mathcal{L} = & - \sum_A (D_\mu \phi)_A^* (D^\mu \phi)_A - \frac{1}{2} \sum_A (\bar{\psi}_A (\not{D} \psi)_A) + \sum_A \mathcal{F}_A^* \mathcal{F}_A \\
& - 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} (\lambda_{AL}^T \epsilon \psi_{CL}) \phi_B^* + i \sum_{ABC} C_{ABC} \phi_B^* \phi_C D_A - \sum_A \xi_A D_A + \frac{1}{2} \sum_A D_A D_A \\
& - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma},
\end{aligned} \tag{27.9.3}$$

where

$$(D_\mu \psi)_A = \partial_\mu \psi_A + \sum_{BC} C_{ABC} V_{B\mu} \psi_C, \tag{27.9.4}$$

$$(D_\mu \lambda)_A = \partial_\mu \lambda_A + \sum_{BC} C_{ABC} V_{B\mu} \lambda_C, \tag{27.9.5}$$

$$(D_\mu \phi)_A = \partial_\mu \phi_A + \sum_{BC} C_{ABC} V_{B\mu} \phi_C, \tag{27.9.6}$$

and

$$f_{A\mu\nu} = \partial_\mu V_{A\nu} - \partial_\nu V_{A\mu} + \sum_{BC} C_{ABC} V_{B\mu} V_{C\nu}. \tag{27.9.7}$$

(Recall that in the adjoint representation $(t_A)_{BC} = -iC_{ABC}$, where C_{ABC} is the real structure constant, defined as usual in this book to include factors of gauge couplings, and taken in a basis in which it is totally antisymmetric.) The Lagrangian density (27.9.3) has $N = 1$ supersymmetry with multiplets $\phi_A, \psi_A, \mathcal{F}_A$ and V_A^μ, λ_A, D_A , because it is a special case of Eq. (27.4.1), and it has an $SU(2)$ symmetry relating ψ_A and λ_A , including invariance under the finite $SU(2)$ transformation (27.9.2), so it also has a *second* independent $N = 1$ supersymmetry with multiplets $\phi_A, \lambda_A, \mathcal{F}_A$ and $V_A^\mu, -\psi_A, D_A$. It therefore satisfies the conditions imposed by $N = 2$ supersymmetry.

We can eliminate the auxiliary fields by setting them equal to the values at which the Lagrangian density (27.9.3) is stationary:

$$\mathcal{F}_A = 0, \quad D_A = -i \sum_{BC} C_{ABC} \phi_B^* \phi_C. \tag{27.9.8}$$

(We are now assuming that the Fayet–Iliopoulos constants ξ_A all vanish.) Inserting these values back in Eq. (27.9.3) gives an equivalent Lagrangian density

$$\begin{aligned}
\mathcal{L} = & - \sum_A (D_\mu \phi)_A^* (D^\mu \phi)_A - \frac{1}{2} \sum_A (\bar{\psi}_A (\not{D} \psi)_A) \\
& + \sqrt{2} \sum_{ABC} C_{ABC} \left(\bar{\psi}_B \frac{1 - \gamma_5}{2} \lambda_A \right) \phi_C - \sqrt{2} \sum_{ABC} C_{ABC} \left(\bar{\lambda}_A \frac{1 + \gamma_5}{2} \psi_C \right) \phi_B^* - V(\phi, \phi^*) \\
& - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma},
\end{aligned} \tag{27.9.9}$$

where the potential is

$$V(\phi, \phi^*) = -\frac{1}{2} \sum_A \left[\sum_{BC} C_{ABC} \phi_B^* \phi_C \right]^2 = 2 \sum_A \left[\sum_{BC} C_{ABC} \operatorname{Re} \phi_B \operatorname{Im} \phi_C \right]^2. \quad (27.9.10)$$

This potential has a minimum value of zero, which is reached not only for $\phi_A = 0$ but also for any set of ϕ s for which $\sum_{BC} C_{ABC} \phi_B^* \phi_C = 0$ for all A or, in other words, for which

$$[t \cdot \operatorname{Re} \phi, t \cdot \operatorname{Im} \phi] = 0, \quad \text{where} \quad t \cdot v \equiv \sum_B t_B v_B. \quad (27.9.11)$$

That is, the minimum of the potential is reached for those scalar fields for which all generators $t \cdot \operatorname{Re} \phi$ and $t \cdot \operatorname{Im} \phi$ belong to a Cartan subalgebra of the full gauge algebra, all of whose generators commute with one another. Though all such values of ϕ give zero potential, and hence unbroken $N = 2$ supersymmetry, they are not physically equivalent, as shown for instance by the different masses they give the gauge bosons associated with the broken gauge symmetries.

One remarkable feature of extended supersymmetry is that the central charges of the supersymmetry algebra in any state can be calculated in terms of ‘charges’ in that state to which bosonic fields are coupled [13]. The easiest way to do this calculation is to use the transformation properties under ordinary $N = 1$ supersymmetry of the extended supersymmetry currents $S_r^\mu(x)$ with $r = 2, 3, \dots, N$ to calculate the anticommutators $\{Q_{1\alpha}, S_{r\beta}^\mu(x)\}$. We can then calculate the central charges from the anticommutators

$$\{Q_{1\alpha}, Q_{r\beta}\} = \int d^3x \{Q_{1\alpha}, S_{r\beta}^0(x)\}. \quad (27.9.12)$$

It turns out that the integrand on the right-hand side is a derivative with respect to space coordinates, but its integral does not vanish if the states have fields that do not vanish rapidly as $\mathbf{x} \rightarrow \infty$.

To see how this works in detail, let’s consider the case of $N = 2$ supersymmetry with an $SU(2)$ gauge symmetry and a single $N = 2$ gauge supermultiplet, with no additional matter supermultiplets. The Lagrangian here is given by Eq. (27.9.3), with A, B , and C running over the values 1, 2, 3, and

$$C_{ABC} = e\epsilon_{ABC}, \quad \xi_A = 0. \quad (27.9.13)$$

(The coupling constant here is denoted e because this is the charge with which the massless gauge field of the unbroken $U(1)$ gauge symmetry interacts.) The usual $N = 1$ supersymmetry current (distinguished now with a subscript 1) is given by Eq. (27.4.40) as

$$\begin{aligned} S_1^\mu = & -\frac{1}{4} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu \lambda_A - e \sum_{ABC} \epsilon_{ABC} \gamma_5 \gamma^\mu \lambda_A \phi_B^* \phi_C \\ & + \frac{1}{\sqrt{2}} \sum_A [(\not{D}\phi)_A \gamma^\mu \psi_{AR} + (\not{D}\phi^*)_A \gamma^\mu \psi_{AL}]. \end{aligned} \quad (27.9.14)$$

We can calculate the second supersymmetry current by subjecting S_1^μ to the finite $SU(2)$ R -symmetry used above, which simply amounts to making the replacements $\psi_A \rightarrow \lambda_A$, $\lambda_A \rightarrow -\psi_A$. This gives

$$S_2^\mu = \frac{1}{4} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu \psi_A + e \sum_{ABC} \epsilon_{ABC} \gamma_5 \gamma^\mu \psi_A \phi_B^* \phi_C + \frac{1}{\sqrt{2}} \sum_A [(\not{D}\phi)_A \gamma^\mu \lambda_{AR} + (\not{D}\phi^*)_A \gamma^\mu \lambda_{AL}] . \quad (27.9.15)$$

It will be enough for our purposes (and somewhat easier) to calculate only the change in the right-handed part of this current under an $N = 1$ supersymmetry transformation. After setting the auxiliary fields equal to their equilibrium values

$$\mathcal{F}_A = 0, \quad D_A = -ie \sum_{BC} \epsilon_{ABC} \phi_B^* \phi_C,$$

we find

$$\begin{aligned} \delta S_{2R}^\mu &= \frac{\sqrt{2}}{4} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu (\not{D}\phi)_A \alpha_R \\ &\quad - \sqrt{2}e \sum_{ABC} \epsilon_{ABC} \gamma^\mu (\not{D}\phi)_A \alpha_R \phi_B^* \phi_C - \frac{\sqrt{2}}{4} f_{A\rho\sigma} (\not{D}\phi)_A \gamma^\mu [\gamma^\rho, \gamma^\sigma] \alpha_R \\ &\quad - \sqrt{2}e \sum_{ABC} \epsilon_{ABC} \phi_B^* \phi_C (\not{D}\phi)_A \gamma^\mu \alpha_R + \dots, \end{aligned}$$

where the dots denote terms bilinear in fermion fields, which do not concern us here because we are interested in effects of long range boson fields. We can combine terms by using the Dirac anticommutation relations and the identity

$$[\gamma^\rho, \gamma^\sigma][\gamma^\mu, \gamma^\nu] + [\gamma^\mu, \gamma^\nu][\gamma^\rho, \gamma^\sigma] = -8\eta^{\mu\rho}\eta^{\nu\sigma} + 8\eta^{\sigma\mu}\eta^{\rho\nu} + 8i\epsilon^{\mu\nu\rho\sigma}\gamma_5$$

and find

$$\begin{aligned} \delta S_{2R}^\mu &= -2\sqrt{2} \sum_A f_A^{\mu\nu} (D_\nu \phi)_A \alpha_R - i\sqrt{2} \sum_A \epsilon^{\mu\nu\rho\sigma} f_{A\rho\sigma} (D_\nu \phi)_A \alpha_R \\ &\quad - 2\sqrt{2}e \sum_{ABC} \epsilon_{ABC} \phi_B^* \phi_C (D^\mu \phi)_A \alpha_R + \dots. \end{aligned}$$

In order to write this as a derivative, we need to use the Yang–Mills field equations given by Eqs. (15.3.6), (15.3.7), and (15.3.9):

$$\begin{aligned} D_\nu f_A^{\mu\nu} &= J_A^\mu = e \sum_{BC} \epsilon_{ABC} ((D^\mu \phi)_B^* \phi_C - \phi_B^* (D^\mu \phi)_C), \\ \epsilon_{\mu\nu\rho\sigma} (D^\nu f^{\rho\sigma})_A &= 0. \end{aligned}$$

These allow us to write δS_{2R}^μ as a total derivative

$$\delta S_{2R}^\mu = D_\nu X^{\mu\nu} \alpha_R, \quad (27.9.16)$$

where

$$X^{\mu\nu} = -2\sqrt{2} \sum_A f_A^{\mu\nu} \phi_A - i\sqrt{2} \sum_A \epsilon^{\mu\nu\rho\sigma} f_{A\rho\sigma} \phi_A + \cdots, \quad (27.9.17)$$

with the dots again indicating irrelevant terms involving fermion fields. Eq. (??) allows us to write Eq. (27.9.16) as an anticommutation relation

$$\{Q_{R\alpha}, S_{R\beta}^\mu\} = i \left[\epsilon \left(\frac{1 - \gamma_5}{2} \right) \right]_{\alpha\beta} D_\nu X^{\mu\nu}. \quad (27.9.18)$$

Since $X^{\mu\nu}$ is a gauge-invariant quantity, its gauge-covariant derivative is the same as its ordinary derivative. Also, $X^{\mu\nu}$ is antisymmetric, so $D_\nu X^{0\nu} = \partial_i X^{0i}$. From Eqs. (27.9.12) and (27.9.18), we have at last

$$\{Q_{R\alpha}, Q_{R\beta}\} = i \left[\epsilon \left(\frac{1 - \gamma_5}{2} \right) \right]_{\alpha\beta} \int dS_i X^{0i}, \quad (27.9.19)$$

with the integral taken over a large closed surface enclosing the system in question, with surface area differential dS taken normal to the surface. Comparing this with Eq. (??) gives the central charge

$$Z_{12} = -i \int dS_i X^{0i}. \quad (27.9.20)$$

If we choose a gauge in which ϕ_A (almost everywhere) has only the constant non-vanishing component $\phi_3 = v$, then

$$\sum_A f_A^{0i} \phi_A = -v E^i, \quad \frac{1}{2} \sum_A \epsilon^{0i\rho\sigma} f_{A\rho\sigma} \phi_A = v B^i, \quad (27.9.21)$$

where $\mathbf{E}$ and $\mathbf{B}$ are the electric and magnetic fields associated with the unbroken $U(1)$ subgroup of the $SU(2)$ gauge group. Therefore the central charge (27.9.20) here is

$$Z_{12} = 2\sqrt{2}v[iq - \mathcal{M}], \quad (27.9.22)$$

where q and $\mathcal{M}$ are the electric charge and magnetic monopole moment, defined by

$$q = \int dS_i E^i, \quad \mathcal{M} = \int dS_i B^i. \quad (27.9.23)$$

As discussed in Section 23.3, this theory, with $SU(2)$ gauge symmetry spontaneously broken by the expectation value of an $SU(2)$ triplet of scalars, is one in which magnetic monopoles actually do occur.

The application of the results of Section 27.4 to the Lagrangian density (27.9.3) shows that, after the spontaneous breaking of the $SU(2)$ gauge symmetry, this theory will contain elementary particles of charge $\pm e$, zero magnetic monopole moment, and tree-approximation mass $M = \sqrt{2}|ev|$. Specifically, for each sign of the charge there is one such particle of spin 1, two of them with spin 1/2, and one with spin 0. It is a striking consequence of the results obtained here that *the mass value $\sqrt{2}|ev|$ is exact, being unaffected by either radiative corrections or non-perturbative effects, provided that the quantity v is defined by Eq. (27.9.22) for the central charge [13].*

To see this, note that the massive one-particle states for each sign of the charge are ‘short’ $N = 2$ supermultiplets, which, as shown at the end of Section 25.5, have masses that saturate the lower bound (??):

$$M = |Z_{12}|/2. \quad (27.9.24)$$

Even if we did not trust the tree approximation to give us the precise value of the particle masses, we would not expect corrections to this approximation to turn short multiplets into the full multiplets, with many more states, that could have larger masses, so we can be confident that Eq. (27.9.24) is exactly valid. For particles with electric charge $q = \pm e$ and zero magnetic monopole moment, Eq. (27.9.20) gives $Z_{12} = \pm 2\sqrt{2}ive$, so Eq. (27.9.24) tells us that their masses are

$$M = \sqrt{2}|ev|. \quad (27.9.25)$$

This is the result found in the tree approximation, but now we see that it is exact.

The semi-classical calculations described in Section 23.3 show that electrically neutral magnetic monopoles in this theory have monopole strengths⁸

$$\mathcal{M} = \frac{4\pi\nu}{e}, \quad (27.9.26)$$

with ν the winding number, a positive or negative integer. The formula (27.9.22) for the central charge together with the inequality (??) thus give a lower bound on the monopole masses

$$M \geq \frac{4\pi\sqrt{2}|\nu v|}{|e|}. \quad (27.9.27)$$

Interestingly, this is the same as the Bogomol’nyi lower bound [14] on monopole energies derived in Section 23.3.⁹ In fact, the monopole solution for $\nu = 1$ that was described in Section 23.3 saturates this bound. More generally, the ‘dyons’ of this theory [15], particles with both charge and magnetic moment, have masses given by [16]

$$M = 2|v|\sqrt{q^2 + \mathcal{M}^2}, \quad (27.9.28)$$

which again is the minimum value allowed by Eqs. (??) and (27.9.20). Indeed, all of the known particles in this theory have masses given in the semi-classical limit by Eq. (27.9.28) [17].

Returning now to general $N = 2$ gauge theories, we may also include additional ‘matter’ fields in the Lagrangian. For simplicity, we will restrict ourselves to ‘short’ massive hypermultiplets (with central charge $\mathcal{Z}$ saturating the inequality (??)), each consisting of a single fermion of spin 1/2 and an $SU(2)$ doublet of spin 0 particles, together with distinct antiparticles. This is the same spin content as is given under $N = 1$ supersymmetry by pairs of left-chiral scalar superfields Φ'_n and Φ''_n , together with their right-chiral adjoints, with the complex scalar field components ϕ'_n and ϕ''_n and their adjoints forming pairs of $SU(2)$ doublets, and the spinor fields all $SU(2)$ singlets. (We are using primes and double

⁸Note that the magnetic moment $\mathcal{M}$ defined by Eq. (27.9.23) is related to the magnetic moment g defined in Section 23.3 by $\mathcal{M} = 4\pi g$.

⁹The canonically normalized field with a non-vanishing vacuum expectation value is $\sqrt{2}\text{Re } \phi_3$ (for real v), so the quantity $\langle \phi \rangle$ appearing in the Bogomol’nyi inequality (23.3.19) is $\sqrt{2}v$.

primes to distinguish these superfields and their components from Φ_A and its components.) If some of these hypermultiplets Φ'_n and Φ''_n are non-neutral under the gauge group, then a superpotential is allowed, of the form:¹⁰

$$f(\Phi, \Phi', \Phi'') = \frac{1}{2} \sum_{Anm} (s_A)_{nm} \Phi'_n \Phi''_m \Phi_A + \frac{1}{2} \sum_{nm} \mu_{nm} \Phi'_n \Phi''_m. \quad (27.9.29)$$

To the Lagrangian density (27.9.3) we then must add a Lagrangian density for these hypermultiplets, given by the first eight terms on the right-hand side of Eq. (27.4.1), and find a total Lagrangian density

$$\begin{aligned} \mathcal{L} = & - \sum_n (D_\mu \phi')_n^* (D^\mu \phi')_n - \sum_n (D_\mu \phi'')_n^* (D^\mu \phi'')_n - \sum_A (D_\mu \phi)_A^* (D^\mu \phi)_A \\ & - \frac{1}{2} \sum_n (\bar{\psi}'_n (\not{D} \psi')_n) - \frac{1}{2} \sum_n (\bar{\psi}''_n (\not{D} \psi'')_n) - \frac{1}{2} \sum_A (\bar{\psi}_A (\not{D} \psi)_A) - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) \\ & + \sum_n \mathcal{F}_n'^* \mathcal{F}'_n + \sum_n \mathcal{F}_n''^* \mathcal{F}''_n + \sum_A \mathcal{F}_A^* \mathcal{F}_A \\ & - \text{Re} \sum_{Anm} (s_A)_{nm} \phi_A (\psi_{nL}'^T \epsilon \psi_{mL}'') - 2\sqrt{2} \text{Re} \sum_{ABC} C_{ABC} (\lambda_{AL}^T \epsilon \psi_{CL}) \phi_B^* \\ & - \text{Re} \sum_{Anm} (s_A)_{nm} \phi'_n (\psi_{mL}''^T \epsilon \psi_{AL}) - \text{Re} \sum_{Anm} (s_A)_{nm} \phi''_m (\psi_{nL}'^T \epsilon \psi_{AL}) \\ & + 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{mn} (\psi_{nL}'^T \epsilon \lambda_{AL}) \phi_m'^* + 2\sqrt{2} \text{Im} \sum_{Anm} (t''_A)_{mn} (\psi_{nL}''^T \epsilon \lambda_{AL}) \phi_m''^* \\ & + \text{Re} \sum_{Anm} (s_A)_{nm} \phi_A \phi'_n \mathcal{F}_m'' + \text{Re} \sum_{Anm} (s_A)_{nm} \phi_A \phi''_m \mathcal{F}_n' + \text{Re} \sum_{Anm} (s_A)_{nm} \phi'_n \phi''_m \mathcal{F}_A \\ & + \text{Re} \sum_{nm} \mu_{nm} \phi'_n \mathcal{F}_m'' + \text{Re} \sum_{nm} \mu_{nm} \phi''_m \mathcal{F}_n' - \text{Re} \sum_{nm} \mu_{nm} (\psi_{nL}'^T \epsilon \psi_{mL}'') \\ & - \sum_{Anm} (t'_A)_{nm} \phi_n'^* \phi_m' D_A - \sum_{Anm} (t''_A)_{nm} \phi_n''^* \phi_m'' D_A + i \sum_{ABC} C_{ABC} \phi_B^* \phi_C D_A \\ & - \sum_A \xi_A D_A + \frac{1}{2} \sum_A D_A D_A - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma}, \end{aligned} \quad (27.9.30)$$

where the $(t'_A)_{nm}$ and $(t''_A)_{nm}$ are the matrices (including coupling constant factors) representing the gauge group on the left-chiral scalar superfields Φ'_n and Φ''_n , respectively. The

¹⁰There still cannot be any terms in the superpotential that are of second or higher order in the Φ_A for the same reason as before: such terms would lead to scalar couplings or masses for the ψ_A , with no corresponding couplings or masses for their $SU(2)$ partners λ_A . Also, there cannot be any term that is trilinear in the Φ'_n and/or Φ''_n , because then the term in Eq. (27.4.1) involving the product of fermion bilinears with second derivatives of the superpotential would lead to a coupling of the $SU(2)$ -singlet fermions to $SU(2)$ doublet fields ϕ'_n or ϕ''_n . Thus the only trilinear interaction terms in the superpotential must involve one factor of Φ_A and two factors of Φ'_n and/or Φ''_n . There cannot be any trilinear terms involving a Φ_A and two Φ'_n s or two Φ''_n s, because that would give the $SU(2)$ singlet auxiliary fields $\mathcal{F}_A$ an interaction with $SU(2)$ -triplet products $\phi'_n \phi'_m$ or $\phi''_n \phi''_m$, and there cannot be any bilinear terms involving two Φ'_n s or two Φ''_n s, because that would yield $SU(2)$ -triplet mass terms $(\psi_{nL}'^T \epsilon \psi_{mL}')^*$ or $(\psi_{nL}''^T \epsilon \psi_{mL}'')^*$. The only remaining allowed bilinear or trilinear terms are of the form (27.9.29).

Yukawa couplings between fermions and scalars have a discrete R -symmetry under the transformation

$$\lambda_{AL} \longrightarrow -\psi_{AL}, \quad \psi_{AL} \longrightarrow +\lambda_{AL}, \quad \phi_n'' \longrightarrow -\phi_n'^*, \quad \phi_n' \longrightarrow \phi_n''^*, \quad (27.9.31)$$

provided that

$$s_A = -2\sqrt{2}i t_A'^T = +2\sqrt{2}i t_A''. \quad (27.9.32)$$

(Note in particular that Eq. (27.9.32) requires the representations of the gauge group furnished by Φ_n' and Φ_n'' to be complex conjugates.) This is also a symmetry of all the other terms in the Lagrangian density (27.9.30), except for those involving the auxiliary fields.

It is not possible to extend the symmetry under the transformation (27.9.31) to the auxiliary fields, but the symmetry appears after the auxiliary fields are eliminated.¹¹ After setting D_A , $\mathcal{F}_n'$, and $\mathcal{F}_n''$ equal to values at which the Lagrangian density is stationary, and combining D - and $\mathcal{F}$ -terms, the Lagrangian density (with s_A and t_A'' given by Eq. (27.9.32),

¹¹After elimination of the auxiliary fields, the resulting action is invariant under the original $N = 2$ supersymmetry transformation only ‘on-shell’—that is, only up to terms that vanish when the fields satisfy the interacting field equations. This doesn’t hurt, because there still are two conserved supersymmetry currents whose integrated time-components satisfy the $N = 2$ supersymmetry anticommutation relations when the fields of which they are composed are required to satisfy the field equations of the Heisenberg picture. ‘Off-shell’ formulations of $N = 2$ supersymmetry exist, but are subject to various complications [18].

and ξ_A taken to vanish) takes the form

$$\begin{aligned}
\mathcal{L} = & - \sum_n (D_\mu \phi')^*_n (D^\mu \phi')_n - \sum_n (D_\mu \phi'')^*_n (D^\mu \phi'')_n - \sum_A (D_\mu \phi)^*_A (D^\mu \phi)_A \\
& - \frac{1}{2} \sum_n (\bar{\psi}'_n (\not{D} \psi')_n) - \frac{1}{2} \sum_n (\bar{\psi}''_n (\not{D} \psi'')_n) - \frac{1}{2} \sum_A (\bar{\psi}_A (\not{D} \psi)_A) - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) \\
& - 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{mn} \phi_A (\psi'^{\text{T}}_{nL} \epsilon \psi''_{mL}) - 2\sqrt{2} \text{Re} \sum_{ABC} C_{ABC} (\lambda^{\text{T}}_{AL} \epsilon \psi_{CL}) \phi^*_B \\
& - 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{mn} \phi'_n (\psi''^{\text{T}}_{mL} \epsilon \psi_{AL}) - 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{mn} \phi''_m (\psi'^{\text{T}}_{nL} \epsilon \psi_{AL}) \\
& + 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{mn} (\psi'^{\text{T}}_{nL} \epsilon \lambda_{AL}) \phi'^*_m - 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{nm} (\psi''^{\text{T}}_{nL} \epsilon \lambda_{AL}) \phi''^*_m \\
& - \frac{1}{4} \sum_A f_{A\mu\nu} f^{\mu\nu}_A + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f^{\mu\nu}_A f^{\rho\sigma}_A \\
& - \sum_{ABnm} \{t'_A, t'_B\}_{mn} \phi_A \phi^*_B (\phi'^*_n \phi'_m + \phi''^*_n \phi''_m) \\
& - \frac{1}{2} \sum_A \left[\sum_{nm} (t'_A)_{nm} (\phi'^*_n \phi'_m - \phi''^*_n \phi''_m) \right]^2 \\
& + \frac{1}{2} \sum_{ABCDE} C_{ABC} C_{ADE} \phi^*_B \phi_C \phi^*_D \phi_E - 2 \sum_A \left| \sum_{nm} (t'_A)_{nm} \phi'_n \phi''_m \right|^2 \\
& - 4 \text{Re} \sum_{nm} (t'_A \mu)_{nm} \phi'^*_n \phi'_m - 4 \text{Re} \sum_{nm} (\mu t'_A)_{nm} \phi''_n \phi''^*_m \\
& - 2 \sum_{nm} (\mu^\dagger \mu)_{nm} \phi'^*_n \phi'_m - 2 \sum_{nm} (\mu \mu^\dagger)_{nm} \phi''_n \phi''^*_m.
\end{aligned} \tag{27.9.33}$$

The last five lines on the right-hand side come from the terms in Eq. (27.9.30) involving auxiliary fields, and now these too are invariant under the discrete transformation (27.9.31), provided that

$$[t'_A, \mu] = [\mu^\dagger, \mu] = 0. \tag{27.9.34}$$

We can now go a step further and consider the case of $N = 4$ extended global supersymmetry. (As remarked in Section 25.4, $N = 3$ supersymmetry is the same as $N = 4$ supersymmetry.) The only massless multiplets of $N = 4$ supersymmetry that do not contain gravitons or gravitinos consist of a single particle of helicity 1, an $SU(4)$ quartet of particles of helicity 1/2, and an $SU(4)$ sextet of particles of helicity 0, together with their CPT-conjugates of opposite helicity. There is one such supermultiplet for each generator t_A of the gauge group. These particles can be grouped into supermultiplets of $N = 2$ supersymmetry: for each t_A there is one gauge supermultiplet consisting of one particle of helicity 1, two particles of helicity $\pm 1/2$, and one particle of helicity 0, together with their CPT-conjugates of opposite helicity, plus two hypermultiplets, each consisting of one particle of each helicity $\pm 1/2$ and two particles of helicity 0. The $N = 2$ gauge superfield consists of an $N = 1$ gauge superfield V_A and a left-chiral scalar superfield Φ_A and its com-

plex conjugate, while the two $N = 2$ hypermultiplets consist of two additional left-chiral scalar superfields Φ'_A and Φ''_A and their complex conjugates.

Since $N = 4$ supersymmetry includes $N = 2$ supersymmetry, the Lagrangian density after elimination of the auxiliary fields of $N = 1$ supersymmetry [18] must be a special case of Eq. (27.9.33), but with the labels n, m , etc. running over the indices A, B, C , etc. of the adjoint representation. Also, the coefficient μ_{nm} in the superpotential (27.9.29) must vanish here, because otherwise Eq. (27.9.33) would contain terms quadratic in the fermion fields ψ'_A and ψ''_A , with no counterpart for their $N = 4$ superpartners λ_A and ψ_A . Also setting $(t'_A)_{BC}$ equal to the generators $-iC_{ABC}$ in the adjoint representation, we find that the Lagrangian density must take the form

$$\begin{aligned}
\mathcal{L} = & - \sum_A (D_\mu \phi')_A^* (D^\mu \phi')_A - \sum_A (D_\mu \phi'')_A^* (D^\mu \phi'')_A - \sum_A (D_\mu \phi)_A^* (D^\mu \phi)_A \\
& - \frac{1}{2} \sum_A (\bar{\psi}'_A (\not{D} \psi')_A) - \frac{1}{2} \sum_A (\bar{\psi}''_A (\not{D} \psi'')_A) - \frac{1}{2} \sum_A (\bar{\psi}_A (\not{D} \psi)_A) - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) \\
& - 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} \phi_A (\psi'^T_{BL} \epsilon \psi''_{CL}) - 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} (\lambda^T_{AL} \epsilon \psi_{CL}) \phi_B^* \\
& - 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} \phi'_B (\psi''^T_{CL} \epsilon \psi_{AL}) - 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} \phi''_C (\psi'^T_{BL} \epsilon \psi_{AL}) \\
& + 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} (\psi'^T_{BL} \epsilon \lambda_{AL}) \phi_C^* + 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} (\psi''^T_{BL} \epsilon \lambda_{AL}) \phi_C^{**} \\
& - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma} - V,
\end{aligned} \tag{27.9.35}$$

where the potential is

$$\begin{aligned}
V = & \sum_{ABCDE} C_{ADE} C_{BCE} (\phi_A \phi_B^* + \phi_B \phi_A^*) (\phi'_C \phi_D'^* + \phi_C'' \phi_D'') \\
& + \frac{1}{2} \sum_A \left| \sum_{BC} C_{ABC} (\phi_B'^* \phi'_C - \phi_B'' \phi_C'') \right|^2 \\
& - \frac{1}{2} \sum_{ABCDE} C_{ABC} C_{ADE} \phi_B^* \phi_C \phi_D^* \phi_E + 2 \sum_A \left| \sum_{BC} C_{ABC} \phi_B' \phi_C'' \right|^2.
\end{aligned} \tag{27.9.36}$$

With no further constraints needed, this Lagrangian has an $SU(4)$ R -symmetry, which implies that it is invariant under $N = 4$ supersymmetry. To see this, we need to use the Jacobi identity to write the cross-term in the second line of the right-hand side of Eq. (27.9.36) in the form

$$\begin{aligned}
\sum_{ABCDE} C_{ABC} C_{ADE} \phi_B'^* \phi'_C \phi_D'' \phi_E'' &= - \sum_{ABCDE} C_{ABC} C_{ADE} \phi_B'^* \phi'_D \phi_E'' \phi_C'' \\
&\quad - \sum_{ABCDE} C_{ABC} C_{ADE} \phi_B'^* \phi'_E \phi_C'' \phi_D'',
\end{aligned}$$

which allows us to write the potential (27.9.36) in a form symmetric among the scalars and their adjoints

$$\begin{aligned}
V = & \sum_A \left| \sum_{BC} C_{ABC} \phi_B^* \phi_C' \right|^2 + \sum_A \left| \sum_{BC} C_{ABC} \phi_B^* \phi_C'' \right|^2 + \sum_A \left| \sum_{BC} C_{ABC} \phi_B \phi_C' \right|^2 \\
& + \sum_A \left| \sum_{BC} C_{ABC} \phi_B \phi_C'' \right|^2 + \sum_A \left| \sum_{BC} C_{ABC} \phi_B'^* \phi_C'' \right|^2 \\
& + \sum_A \left| \sum_{BC} C_{ABC} \phi_B' \phi_C'' \right|^2 + \frac{1}{2} \sum_A \left| \sum_{BC} C_{ABC} \phi_B' \phi_C'^* \right|^2 \\
& + \frac{1}{2} \sum_A \left| \sum_{BC} C_{ABC} \phi_B'' \phi_C''^* \right|^2 + \frac{1}{2} \sum_A \left| \sum_{BC} C_{ABC} \phi_B \phi_C^* \right|^2.
\end{aligned} \tag{27.9.37}$$

Now to make the $SU(4)$ symmetry apparent, we introduce an $SU(4)$ notation for the fields. We assemble the left-handed fermion fields into an $SU(4)$ vector:

$$\psi_{1AL} \equiv \psi_{AL}, \quad \psi_{2AL} \equiv \lambda_{AL}, \quad \psi_{3AL} \equiv \psi'_{AL}, \quad \psi_{4AL} \equiv \psi''_{AL}. \tag{27.9.38}$$

In order for the fermion kinematic terms in the Lagrangian density to be $SU(4)$ -invariant, we then must assemble the right-handed fermion fields into a contragredient vector:

$$\psi_{AR}^1 \equiv \psi_{AR}, \quad \psi_{AR}^2 \equiv \lambda_{AR}, \quad \psi_{AR}^3 \equiv \psi'_{AR}, \quad \psi_{AR}^4 \equiv \psi''_{AR}. \tag{27.9.39}$$

The Majorana condition on the fermion fields then takes the $SU(4)$ -invariant form

$$(\psi_{iAL})^* = -\beta \epsilon \psi_{AR}^i, \tag{27.9.40}$$

with indices i, j , etc. running over the values 1, 2, 3, 4. In order for the Yukawa couplings between the fermion and scalar fields to be $SU(4)$ -invariant, we must give the scalars the transformation properties of an antisymmetric $SU(4)$ tensor

$$\begin{aligned}
\phi_A^{12} &\equiv \phi_A^*, & \phi_A^{13} &\equiv \phi_A'', & \phi_A^{14} &\equiv -\phi_A', \\
\phi_A^{23} &\equiv -\phi_A'^*, & \phi_A^{24} &\equiv -\phi_A''^*, & \phi_A^{34} &\equiv \phi_A,
\end{aligned} \tag{27.9.41}$$

which also obeys an $SU(4)$ -invariant reality condition

$$(\phi_A^{ij})^* = \frac{1}{2} \sum_{kl} \epsilon_{ijkl} \phi_A^{kl}. \tag{27.9.42}$$

The whole Lagrangian density (27.9.35) can then be written in the manifestly $SU(4)$ -invariant form

$$\begin{aligned}
\mathcal{L} = & -\frac{1}{2} \sum_{Aij} (D_\mu \phi^{ij})_A (D^\mu \phi^{ij})_A^* \\
& -\frac{1}{2} \sum_{Ai} (\psi_{iAL}^T \epsilon (\not{D} \psi_R^i)_A) + \frac{1}{2} \sum_{Ai} (\psi_{AR}^{iT} \epsilon (\not{D} \psi_{iL})_A) \\
& -\sqrt{2} \operatorname{Re} \sum_{ABCij} C_{ABC} \phi_A^{ij} (\psi_{iBL}^T \epsilon \psi_{jCL}) - V \\
& -\frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma},
\end{aligned} \tag{27.9.43}$$

where the potential is

$$V = \frac{1}{8} \sum_{Aijkl} \left| \sum_{BC} C_{ABC} \phi_B^{ij} \phi_C^{kl} \right|^2. \quad (27.9.44)$$

The potential has a minimum value zero, so that supersymmetry is not broken in this theory. The minimum is reached when the generators $\sum_A t_A \phi_A^{ij}$ all commute with one another.

For zero theta angle, the gauge theories with a simple gauge group and either $N = 2$ or $N = 4$ supersymmetry have just a single coupling constant, the gauge coupling constant g . Since these theories have $N = 1$ supersymmetry, they share the property discussed in Section 27.6, that the only infinity in higher orders of perturbation theory is in a one-loop correction to this coupling.¹² The function $\beta(g)$ in the renormalization group equation $\mu dg/d\mu = \beta(g)$ is then given to all orders of perturbation theory by the one-loop formula (18.7.2), with a suitable correction for the presence of scalar fields:

$$\beta(g) = -\frac{g^3}{4\pi^2} \left(\frac{11}{12} C_1 - \frac{1}{6} C_2^f - \frac{1}{12} C_2^s \right), \quad (27.9.45)$$

where

$$\begin{aligned} \sum_{AB} C_{ABC} C_{ABD} &= g^2 C_1 \delta_{CD}, \\ [\text{Tr}(t_C t_D)]_{\text{Majorana fermions}} &= g^2 C_2^f \delta_{CD}, \\ [\text{Tr}(t_C t_D)]_{\text{complex scalars}} &= g^2 C_2^s \delta_{CD}. \end{aligned} \quad (27.9.46)$$

In general theories with $N = 2$ supersymmetry we have two Majorana fermions λ_A and ψ_A in the adjoint representation and H pairs of Majorana fermions ψ'_n and ψ''_n whose left- and right-handed parts are in representations with generators either t'_A or $-t_A^T$, so

$$C_2^f = 2C_1 + 2HC'_2, \quad (27.9.47)$$

where C'_2 is defined by

$$\text{Tr } t'_C t'_D = g^2 C'_2 \delta_{CD}. \quad (27.9.48)$$

Also, we have one complex scalar ϕ_A in the adjoint representation and H pairs of complex scalars ϕ'_n and ϕ''_n in representations with generators t'_A or $-t_A^T$, so

$$C_2^s = C_1 + 2HC'_2. \quad (27.9.49)$$

The beta function (27.9.45) is therefore

$$\beta(g) = -\frac{g^2}{8\pi^2} (C_1 - HC'_2). \quad (27.9.50)$$

The case of $N = 4$ supersymmetry is just the special case with $H = 1$ pairs of $N = 2$ hypermultiplets in the adjoint representation, with $C'_2 = C_1$, so in this case the beta function vanishes. This is therefore a finite theory, with no renormalizations at all [19].

¹²The trilinear term in the superpotential (27.9.29) is proportional to the gauge coupling, and is therefore renormalized, despite the no-renormalization theorem of Section 27.6. This is because here we are renormalizing the left-chiral scalar superfields Φ_A, Φ'_n , and Φ''_n as well as the gauge superfield V_A so as to keep them canonically normalized. The bilinear term in Eq. (27.9.29) is renormalized for the same reason.

Gauge theories with $N = 4$ supersymmetry have another remarkable property, known as *duality*. This was first conjectured by Montonen and Olive [17] for purely bosonic theories in which a simple gauge group is spontaneously broken to a $U(1)$ electromagnetic gauge group. They noticed that semi-classical calculations (of the sort described in Section 23.3) give the mass of particles with charge $q = ne$ and magnetic monopole moment $\mathcal{M} = 4\pi m/e$ (with n and m integers of any sign) as

$$M = \sqrt{2} \left| v \left(ne + \frac{4\pi im}{e} \right) \right|, \quad (27.9.51)$$

which is invariant under the transformations

$$m \longrightarrow n, \quad n \longrightarrow -m, \quad e \longleftrightarrow 4\pi/e. \quad (27.9.52)$$

On this basis, they suggested that the theory with a weak gauge coupling e is fully equivalent to one with a strong gauge coupling $4\pi/e$. Neither the purely bosonic theory nor the simplest versions of the $N = 1$ and $N = 2$ extended supersymmetry theories really have this property; [20] for one thing, the massive charged elementary vector bosons of the broken gauge symmetries have spin 1, while all of the monopoles and dyons have spins 1/2 or 0. (We will see in Section 29.5 that $N = 2$ theories do have a duality property of a more subtle sort.) But for $N = 4$ supersymmetry the monopole states form multiplets with one particle of spin 1, four particles of spin 1/2, and two particles of spin 0, just like the elementary particles [20]. Evidence has accumulated [21] that $N = 4$ supersymmetric gauge theories are indeed invariant under the interchange of electric and magnetic quantum numbers and of e with $4\pi/e$. The equivalence of theories with large and small coupling constants has become an increasingly important theme in string theory, but this is beyond the scope of this book.

Problems

1. To second order in gauge coupling constants, calculate the components of the superfields Ω^A that are needed to put the gauge superfield V^A in Wess–Zumino gauge by the transformation (27.1.12).
2. Show that the most general chiral spinor superfield W_α satisfying the condition (27.2.20) has components $f_{\mu\nu}$ satisfying the homogeneous Maxwell equations $\epsilon^{\mu\nu\rho\sigma} \partial_\rho f_{\mu\nu} = 0$. What conditions on the other components of W_α are imposed by Eq. (27.2.20)?
3. Consider a general renormalizable $N = 1$ supersymmetric gauge theory with an $SU(2)$ gauge group and a single chiral superfield belonging to the 3-vector representation of $SU(2)$. What is the most general superpotential for this theory? Construct the Lagrangian density of the whole theory explicitly. Eliminate the auxiliary fields. Show that supersymmetry is not broken in this theory. What are the masses of the particles of this theory?
4. Express the gaugino and chiral fermion fields in the supersymmetric version of quantum electrodynamics described in Section 27.5 in terms of the goldstino field and other spinor fields of definite mass.

5. Consider the renormalizable $N = 2$ supersymmetric theory with an $SU(3)$ gauge symmetry and no hypermultiplets. What are the values of the scalar fields for which the potential vanishes? What are the massless gauge fields for non-zero values of these scalars? Calculate the central charge in terms of the quantities to which these massless gauge fields are coupled.

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